Beyond Boundaries:

Utilizing Space Syntax to Reconsider Conservation Area Appraisals and Policy in Suburban London

by

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Abstract

This thesis examines the extent to which conservation area policy in Greater London describes the aspects of spatial morphology that underpin the urban vitality of the areas designated. Previous research has established that urban vitality depends on spatial complexity, and that the erosion of spatial complexity can have detrimental effects on existing urban vitality. Those who designate conservation areas, especially those encompassing town centers and high streets, therefore have an interest in conserving the network characteristics of space produced by morphological complexity that sustain urban dynamism.

This thesis explores this by assessing the “spatial language” of the local plans and conservation areas of two suburban London town centers: South Norwood and Teddington. In the absence of more specific urban conservation frameworks in England, local plans and conservation area appraisals are the most appropriate vehicle for network-morphological conservation, given their recognition of the importance of heritage and their use of the National Planning Policy Framework’s definition of “setting,”—a designation that is protected yet flexible enough to accommodate descriptions beyond viewpoints and vistas to which it has traditionally applied.

This thesis uses space syntax to produce novel QGIS-based visualizations of segment angular analysis for choice and integration at multiple movement scales in the Teddington study area. These visualizations, which proxy spatial potential for urban activity, are mapped against the distribution of conservation areas around Teddington to assess alignment between area boundary logic and movement patterns sustaining urban activity.

The results reveal weakness in the spatial descriptions found in conservation area appraisals. Rarely do appraisals account for movement dynamics, building-to-street relationships, built form to land use relationships, or connectivity. The placement and justification of area boundaries, when compared to maps of spatial potential, fail to account for the geographic scope of the networks within which they operate. This thesis argues that the Teddington, and other areas like it, face long-term risk of isolation and further fragmentation, and that current conservation area appraisals remain largely focused on aesthetic and experiential qualities of the built environment.

This research adds to the limited number of studies combining heritage and space syntax methodology, and provides the basis for a framework to reassess conservation policy so that it operates in closer alignment with emerging global frameworks. These frameworks intend to unify urban planning, spatial research, and heritage study to prevent historic areas from dissolving physically, economically, or culturally as a result of being frozen in time.

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I'm probably a lot more closer in the 1500m
to the world record than I am in the 5000m.

– Mo Farah, Teddington native, on
different scales of movement

1. Introduction

As London continues to grow, planning frameworks have been developed to address the complex goal of fostering resilient and vital communities. Often these planning goals go hand in hand, but also head to head, with the city's historical urban fabric. It is generally accepted that historic value contributes to the character, desirability, and vitality of communities.[1,2] In response, English civic authorities have authored numerous frameworks and tools meant to sustain and integrate areas of heritage value within growing urban localities.

Since the Civic Amenities Act of 1967, one such tool for local governments is the ability to designate conservation areas that are meant to protect the historic and cultural aspects of areas valuable to residents and visitors. The guidelines for said appraisals have also become more holistic as urban studies have advanced them beyond appraisals of aesthetic value, and stressed consideration of the entire townscape. This can include streetscapes, views, connectivity, and accessibility. This collection usually falls under the umbrella of "setting." The scope and scale of these designated areas varies greatly across Greater London, and each is separately appraised and managed by separate local planning authorities within each borough.

Designated conservation areas are often high streets and town centers. These areas are the hearts of communities in a historical, cultural, and economic sense. They are frequently the oldest, most aesthetically interesting, and hold the "je ne sais quoi" of the local urban character.

[1] United States Conference of Mayors, Special Committee on Historic Preservation et al., *With Heritage So Rich: A Report of a Special Committee on Historic Preservation*, ed. National Trust for Historic Preservation (Random House, 1966; repr., Preservation Books, 1999), 7.
[2] Nicola Dempsey, "Quality of the Built Environment in Urban Neighbourhoods," *Planning Practice and Research* 23, no. 2 (2008): 249–64, https://doi.org/10.1080/02697450802327198.

On the other hand, the *active* role of space (as opposed to the experiential role) has been explored in depth both qualitatively and quantitatively for decades under disparate academic umbrellas. Seminal 20th century scholarship broadly described the emergence of vitality in urban environments from the perspective of on-the-ground observers, detailing the features of the built environment that seemed to contribute to movement, community, and the effective use of space. The quantitative branch of study "space syntax" emerged in the late 1970s, attempting to explain how space and movement mutually structure each other, and rationalizing the emergent patterns of urban spatial complexity that can be seen in cities throughout the world.

Beginning with Bill Hillier and Julienne Hanson's foundational works,[3,4] space syntax has since been used to study the configurational properties of space and its effects on the distribution of people and vitality of communities. However, despite the number of studies that adopt a historical approach to aid in these explanations, it has also been recognized that there is very little overlap between space syntax and historic conservation and heritage studies, despite the potential.[5]

This thesis strives for a cross-disciplinary approach to fill this gap. It interrogates the relationship between the definitions of specific conservation area boundaries and the descriptions of space therein, and the structuring role of space and its capacity to seed the activities that give these spaces life.

Chapter two is a review of existing scholarship and policy relevant to urban morphology and conservation. The first section examines the literature which has described the ways in which morphological complexity and network connectivity are essential aspects of maintaining urban resiliency. The second section provides an overview of UK policy and guidance, and global frameworks. Chapter three illustrates the methodology by which the London suburban town center case studies were assessed. Chapter four analyzes the results of the quantitative and qualitative evaluations and investigates the

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[3] Bill Hillier, *Space Is the Machine: A Configurational Theory of Architecture* (Press Syndicate of the University of Cambridge, 1996; Electronic Edition, Space Syntax, 2007).
[4] Bill Hillier and Julienne Hanson, *The Social Logic of Space* (Cambridge University Press, 1984; Online Edition, Cambridge University Press, 2009).
[5] Garyfalia Palaiologou and Sam Griffiths, "The Uses of Space Syntax Historical Research for Policy Development in Heritage Urbanism," in *Cultural Urban Heritage*, ed. Mladen Obad Šćitaroci et al., The Urban Book Series (Springer International Publishing, 2019), 19–34, https://doi.org/10.1007/978-3-030-10612-6_2.

degree to which conservation area boundaries and protection of their respective settings is sufficient for protecting these aspects. The final chapter summarizes these findings and presents an argument that urban complexity is itself a heritage asset worthy of inclusion in conservation and planning policy.

This thesis intends to be a small step towards broadening English conservation bodies' understanding of conservatorship and the resulting policies; and introduce the idea that the vitality within heritage areas is more deeply connected to the surrounding spatial fabric than previously recognized. This in turn has implications for future policy recommendations and planning frameworks produced by the UK's historic conservation bodies and urban planning departments, as well as increasing the visibility of space syntax in global conservation dialogue.

2. Background

2.1 The Agency of Space

Lest historic cities be reduced to historic ruins, it is paramount that these places maintain their urban vitality. Although the debate as to what formally defines urban vitality is still ongoing, this thesis interprets it as the visible activity of people engaging in a variety of social, cultural, and economic activities, in multiple places, and at a variety of times. This notion of urban complexity, and its faltering in mid-20th century New York, is what Jane Jacobs seeks to understand in the seminal work *Death and Life of Great American Cities*.[6] In doing so, Jacobs describes the qualities of urban spatial morphology that amount to urban vitality. In the city, buildings must have a mixture of primary uses to support the needs and activities of many people in shared space. There must be small blocks featuring multiple intersections to allow for ease of movement at different scales and multiple businesses hosted in small areas. There must be a mixture of old and new buildings for, without even an explicit consideration for heritage, older buildings are cheaper and can be generators for new business activity. And cities need a good density of

[6] Jane Jacobs, *Death and Life of Great American Cities* (Random House, 1961; Vintage Books Edition, Vintage Books, 1992).

people to support this activity and each other.[7] These are the spatial qualities that allow a city to generate a mix of activities, allow people to move to and from those activities, and support a sense of community and safety–what Jacobs calls the "ballet of the sidewalk."[8]

It is worth noting how much of what Jacobs describes is at least partially reliant on the urban grid of streets, and although Jacobs might decry formal efforts towards top-down city planning, there are numerous examples where semi-planned grids (including New York City) have generated the organic growth and physical qualities of urban vitality that Jacobs observes.[9]

The power of the street network as an organizing tool is an essential aspect of urban morphological research and a key aspect of the case studies presented in this thesis. Although not addressed statistically until a decade or so later, the agency the grid has in determining where and how people move throughout the urban environment was well described by Kevin Lynch in *Image of the City*, where he argues "the paths, the network of habitual or potential lines of movement through the urban complex, are the most potent means by which the whole can be ordered."[10] It is the street network that provides the substrate for the arrangement of buildings, uses, and forms that generate the necessary movement to sustain urban life. Lynch also argues that there would be a lack of urban activity if the city was not intelligible; and the street network is critical for the mental maps that piece together the unique physical references at the micro and macro level that allow for people to move to and from their needs and desires confidently at a variety of scales.

Ultimately, what Jacobs and Lynch are describing is urban complexity, a varied array of morphologies constituting an urban network supporting a diverse range of activity. Urban complexity's relationship to urban vitality has long been investigated by historical and spatial researchers, and it will be the focal point of this thesis going forward.

[7] Jacobs, *Death and Life of Great American Cities*, 150.
[8] Jacobs, *Death and Life of Great American Cities*, 50.
[9] Leslie Martin, "The Grid as Generator," in *Urban Space and Structures*, ed. Leslie Martin and Lionel March (Cambridge University Press, 1972; repr., Cambridge University Press, 1972), 7.
[10] Kevin Lynch, *The Image of the City*, Publication of the Joint Center for Urban Studies (1960; 33. print, M.I.T. Press, 2008), 96.

2.2 Space Syntax

Space syntax is a branch of spatial research that has sought to further investigate urban complexity by mathematically describing network properties of space, and how these properties give rise to the observed qualities of a successful and lively urban environment. As stated previously, the work of Bill Hillier and Julienne Hanson described the way in which space and movement mutually structure each other. Stated simply, throughout history there is a cycle in which people structure space physically, and the physical form of that space has an active role in how people move throughout it, and where they are likely to conduct different activities as a result of this movement. This in turn leads to actions concerning the continual expansion and interconnectivity of space, and so forth.

Space syntax adds a quantitative element to this notion. It assigns values to streets based on their placement in relation to other parts of the network, describing the degree to which each street will affect movement patterns. The underlying notion is that space is *generative* of movement and space syntax can provide statistical approximations of where people are likely to be at a given time,[11] and which routes of connectivity have the highest importance in a network.

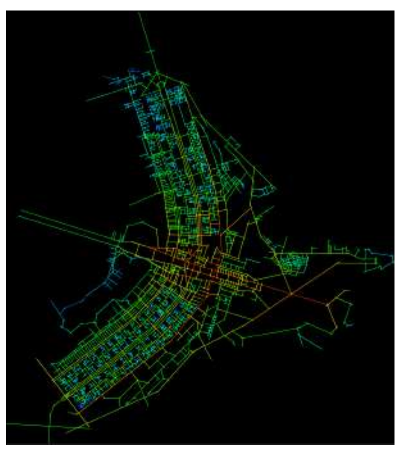

Figure 1
Map of axial lines in Brasília. The colours show the global integration of the different streets, measuring the accessibility of a topological line for the entire system according to the spatial analysis of the space syntax. Created with Mindwalk 1.0
Source: Tony Rotondas - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=5403041

[11] Bill Hillier, "Centrality as a Process: Accounting for Attraction Inequalities in Deformed Grids," *Proceedings of the 2nd International Space Syntax Symposium* 2 (1999): 2.

Two ‘syntactic’ measures common in space syntax research are *choice* and *integration*, which function as proxies for *through-movement*—the routes which people will use most often to get from point A to point B—and *to-movement*—routes which are themselves often the end points of local trips given their relative centrality in a network. Mapping these measures creates images akin to heatmaps visualizing which street segments have higher value, or greater importance, in the network (see figure 1). These representations can also map where there is likely to be “co-presence,” places where people are likely to encounter each other as their patterns of movement and trip destinations overlap. In these locations, urban morphology is likely to seed “live activity,”[12] or businesses that establish active frontages to capture the economic potential of high numbers of people. This usually includes retail and commercial-communal spaces such as bars, cafes, and restaurants.

However, in the context of small town centers that this thesis focuses on, “live activity” is not self-sustaining and relies on the greater urban complexity to generate the movement it depends on. Town centers and high streets are often surrounded by a network of background economic activity that equally takes advantage of what the space provides. Small scale storage and industrial spaces tend to stick to back streets and turn-offs from main thoroughfares. And schools, leisure facilities, offices, and medical and professional services are scattered throughout the accessible reach of town centers.[13] This larger and dynamic “active center” is the emphasis of Laura Vaughan’s work, which uses space syntax methodology to describe the relationship between morphological complexity and resiliency—vitality over time—in small town centers.[14] The “spatial signature” Vaughan observes in small town centers demonstrates the ability to foster co-presence at a variety of movement scales. This means that co-presence occurs at the numerous spatial intersections of residents and visitors going on different trips, for different purposes, of different lengths. This in turn is related to the capacity of the town to support an array of mixed uses that

[12] Hillier, “Centrality as a Process: Accounting for Attraction Inequalities in Deformed Grids,” 1; Laura Vaughan et al., “Beyond the Suburban High Street Cliché - A Study of Adaptation to Change in London’s Street Network: 1880-2013,” *Journal of Space Syntax* 4, no. 2 (2013): 238; Julienne Hanson, “Urban Transformations: A History of Design Ideas,” *URBAN DESIGN International* (UK) 5 (2000): 120.
[13] Francesca Froy, *Rebuilding Urban Complexity: A Configurational Approach to Postindustrial Cities*, 1st ed. (Routledge, 2024), 39, https://doi.org/10.4324/9781003349990.
[14] L. Vaughan et al., “The Spatial Signature of Suburban Town Centres,” pt. 77-91, *Journal of Space Syntax* 1, no. 1 (2010): 77.

go beyond retail. Furthermore, the historic built environment is composed of an array of forms that are flexible to changing use, even when those use-categories change over time[15] Vaughan validates Jacobs' observations of urban vitality at a given place at a given time by describing the morphological complexities that set the stage for the sidewalk ballet.

Although it may seem clear that urban complexity is a critical aspect of lasting urban adaptability, and thus vitality, can it be said that a lack of urban complexity undermines a vital community? In one study, Vaughan also examines the effect of diminished urban complexity. In the case of South Norwood (one of the case studies re-examined in this thesis), Vaughan notes that the urban grid's intensification is visibly disrupted by the incursion of railway and major road networks. As a result, the capacity to support co-presence at a variety of scales, as opposed to only one form of movement directive, has been diminished, resulting in a shrinking of the local economy, especially in relation to other towns in her case studies.[16]

2.3 Urban Planning and Ruptured Urban Fabrics

Estate plans, garden cities, and suburbification have been excoriated over multiple decades. Anthony Tung's historical account of conservation practice describes early estate planning as well-landscaped but ultimately commercially driven attempts to segregate by social class.[17] These estates had limited connectivity and were sustained by amassed wealth which had little interest in the vitality of urban life. When the same suburban ideal was applied to mass-housing projects à la Le Corbusier and subsequent modernist thinkers, Garden City enthusiasts, or "City Beautiful" diehards, the results were isolated, impoverished, and unsafe cities that failed to support the livelihoods of their residents. Jacobs colorfully laments these attempts to "urban plan" and notes that ultimately these morphologies attempt to

[15] Laura Vaughan et al., "An Ecology of the Suburban Hedgerow, or: How High Streets Foster Diversity over Time," *Proceedings of the 10th International Space Syntax Symposium*, 2015, 16.
[16] Vaughan et al., "An Ecology of the Suburban Hedgerow, or: How High Streets Foster Diversity over Time," 9–11.
[17] Anthony M. Tung, "Chapter 10: The Comprehensible Urban Visage," in *Preserving the World's Great Cities: The Destruction and Renewal of the Historic Metropolis* (Three Rivers Press, 2001), 283–84.

remove urban processes from urban land and replace them with sentimental notions of the world outside the city.[18]

Revisiting the common theme of quantitative and qualitative spatial research to validate the ever-present Jacobs, Julienne Hanson directs attention to the development and redevelopment of Somers Town in London. Using numerous syntactic measures to trace the development of Somers Town, she documents the shift from outward-facing development—with streets constituted by building entrances and density-maximizing design—to inward-facing, fragmented, and bounded morphologies. Even where overall building shape and plot size remained consistent with regard to the grid, the shift of outward- to inward-facing spatial organization had a detrimental effect on mobility and social cohesion due to the destruction of morphological complexity.[19]

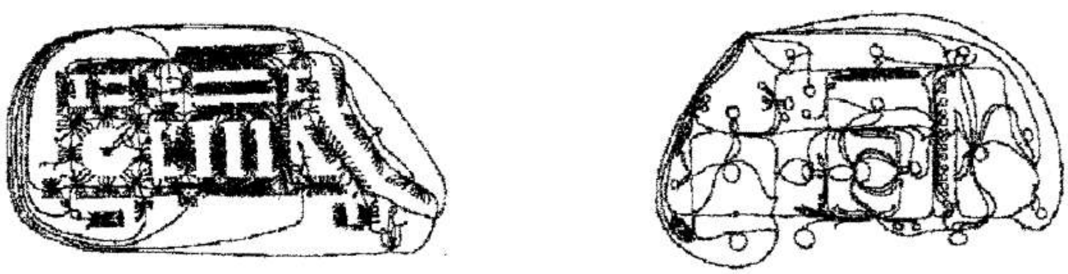

Figure 2a-b:
Interface maps of 19th century (left) and 20th century (right) Somers Town demonstrating how new urban development resulted in a decline in the rate at which buildings "constitute" streets, i.e. fewer front facing entrances and more unused bounded space. The takeaway here is a visual representation of how 20th century development schemes were inward and attempted to pre-program and bind space in a way that led to measurable breakdown of community vitality as could be described by Jacobs (1961) and expanded upon by Hanson (2000).

Source: Hanson, "Urban Transformations: A History of Design Ideas," 106.

The "ruptured interface" of Somers Town (of which there have been more recent attempts to repair), meant fewer neighbors, fewer eyes on the street, fewer economic functions, and more 'dead space.' Instead of space being host to numerous behaviors, space is designed to dictate and control behavior to the whims of those least likely to inhabit these spaces at all.

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[18] Jacobs, *Death and Life of Great American Cities*, 443–48.
[19] Hanson, "Urban Transformations: A History of Design Ideas," 100–117.

In perhaps the most comprehensive review, Francesca Froy's *Rebuilding Urban Complexity*[20] examines the case of Manchester and other post-industrial cities' troubled histories with post-war development. Manchester, as with the other case studies, grew due to intellectual and economic exchange facilitated by its organic urban form. But its regional evolution takes different shapes with the exit of major industries. In some areas, the incursion of estate-based plans, business parks, raised highways, large roundabouts, larger lots, and fewer instances of local mixed use amounts to a reduction of urban complexity, movement capacity, and economic potential. In other areas, such as the northern quarter, Froy celebrates the "enduring spatial potential"[21] where the "griddy" and integrated morphology has facilitated frequent re-use and transformation of this area; from textiles, to artisans, to chic retail-focused communities in the present day.

When asked if urban morphology should be considered a heritage asset, Froy offers a resounding "yes."[22] This thesis builds on the conjecture that urban complexity and connectivity is a critical feature of space worth understanding qualitatively and quantitatively, and that steps to conserve and enhance it are critical for urban resilience. The following chapter of this review focuses on the policies in place for planning and heritage that ostensibly work to make this possible.

2.4 The Hierarchy of English Urban Policy

The spatial complexity at all urban scales is not just the result of social or economic processes. Nor is it purely dictated by spatial form. The development of urban complexity is also a historical process, and thus an area's heritage value is ultimately bound up in the complexity of its morphological relationships. While 'urban complexity' itself might be too broad to protect on its own, a fuller accounting of the *active role* of space, beyond the experiential one, can provide a new framework for informing future heritage policy.

[20] Froy, *Rebuilding Urban Complexity*.
[21] Francesca Froy, "Conversation with Dr. Francesca Froy," (Paris, France), Alec J. Ranzato, February 25, 2026, Video Conference.
[22] Froy, "Conversation with Dr. Francesca Froy."

This thesis focuses on the strengths and limitations of heritage conservation policy in London, specifically the use, scope, and effectiveness of conservation areas. By revisiting a case study used by Vaughan and contributing an additional case study unique to this thesis, the scope further focuses on the relationship between urban complexity at the London-suburb level, and conservation areas that address these specific types of organization.

However, an effective explanation of the use, scope, and effectiveness of designated conservation areas necessitates a prior investigation of the hierarchy of urban policy in the UK that activates and manages these areas.

2.4.1 Conservation Areas

The Civic Amenities Act of 1967 was the first law passed to both define what conservation areas are, and give local councils the authority to designate them and approve planning applications that affect these areas. The language of the act is as follows stating that conservation areas are:

> *"areas of special architectural or historic interest the character or appearance of which it is desirable to preserve or enhance, and shall designate such areas (hereafter referred to as Conservation Areas ") for the purpose of this section."*[23]

The statutory management of conservation areas, and listed buildings, was further expanded in the Planning (Listed Buildings and Conservation Areas) Act 1990, which remains the basis of local planning authority over heritage assets under their jurisdiction. Given that this thesis considers an expanded definition of "setting" and its protections, the relevant areas of PLBCAA are sections S16(2), S66(1), S69, and S72(1). These sections collectively state that both listed building consent and planning permissions granted by local planning authorities must consider the "desirability of preserving the building *or its setting*,"[24] granting statutory protections to the setting of heritage assets.

[23] *Civic Amenities Act 1967*, c. 69 (UK).
[24] *Planning (Listed Buildings and Conservation Areas) Act 1990*, c. 9 (UK), s. 16(2).

What the act notably does *not* do is protect the setting of conservation areas themselves. Instead, it directs special attention to "preserving or enhancing the character or appearance of that area,"[25] in conjunction with regular review and reconsideration of these areas.

2.4.2 The National Planning Policy Framework

The definition and protection of setting, with regard to conservation areas, actually trickles down from guidance via the National Planning Policy Framework.[26] Although periodically reviewed and rewritten, its current form and the definitions therein are from the 2024 iteration (with a new version anticipated in 2026 by the time of this thesis' review).

The NPPF is an extensive set of guidelines meant to direct planning policy and advise local councils on how best to achieve the political livability aims of the current administration. It sets out a number of goals, and the one relevant here is the tie between sustainable urban development, the vitality of town centers, and the conservation of the historic environment, as described in sections 2, 7, and 16 respectively.

The framework emphasizes the importance of local town centers, while also defining their boundaries, and urging planning authorities to consider their connectivity, accessibility, and rich mix of uses. Additionally, when considering the protection of heritage assets, which are often coextensive with town centers, these "irreplaceable resource[s]" should be considered within the "wider social, cultural, economic and environmental benefits that conservation of the historic environment can bring."[27] It is in this section that conservation areas are themselves validated as heritage assets, with their settings likewise subject to appraisal and protection.

[25] *Planning (Listed Buildings and Conservation Areas) Act 1990*, c. 9 (UK), s. 77(1).
[26] Ministry of Housing, Communities & Local Governments, *National Planning Policy Framework*, (UK), 2024.
[27] Ministry of Housing, Communities & Local Governments, *National Planning Policy Framework*, para. 203(e).

The following NPPF definitions are relevant to this thesis:

Designated heritage asset	A World Heritage Site, Scheduled Monument, Listed Building, Protected Wreck Site, Registered Park and Garden, Registered Battlefield or Conservation Area designated under the relevant legislation.
Heritage asset	A building, monument, site, place, area or landscape identified as having a degree of significance meriting consideration in planning decisions, because of its heritage interest. It includes designated heritage assets and assets identified by the local planning authority (including local listing).
Setting of a heritage asset	The surroundings in which a heritage asset is experienced. Its extent is not fixed and may change as the asset and its surroundings evolve. Elements of a setting may make a positive or negative contribution to the significance of an asset, may affect the ability to appreciate that significance or may be neutral.
Town center	Area defined on the local authority's policies map, including the primary shopping area and areas predominantly occupied by main town center uses within or adjacent to the primary shopping area. References to town centers or centers apply to city centers, town centers, district centers and local centers but exclude small parades of shops of purely neighbourhood significance. Unless they are identified as centers in the development plan, existing out-of-center developments, comprising or including main town center uses, do not constitute town centers.

Notably, the NPPF's definition of 'setting' does not make any specific reference to spatial aspects such as connections, streetscapes, and interbuilding relationships, but some of this is expanded upon in the further guiding framework provided by Historic England in *The Setting of Heritage Assets*.[28]

Furthermore, the NPPF is rather conservative in its recommendations to local councils in their ascription of heritage and heritage boundaries, urging for a thorough review of breadth and scale of conservation areas lest the notion of heritage value be diluted.[29] However, as a document that considers the value of urban heritage and includes it within the larger discourse of urban vitality and urban form, it is in alignment with the developments in global conservation.

[28] Historic England, *The Setting of Heritage Assets: Historic Environment Good Practice Advice in Planning Note 3 (Second Edition)* (Historic England, 2017).
[29] Ministry of Housing, Communities & Local Governments, *National Planning Policy Framework*, para. 204.

2.4.3 UNESCO and The Historic Urban Landscape Approach

UNESCO guidance has begun to consider the entirety of an historic urban area's stratigraphy in a physical, cultural, and temporal sense. This has been termed the Historic Urban Landscape (HUL) approach. In a *Recommendation on the Historic Urban Landscape*[30] UNESCO recognizes the need to better integrate conservation and heritage work into the larger urban context, stating explicitly that heritage conservation has to move beyond isolated examples of architectural and cultural value and consider what this work has to offer towards sustainable urban development and the improvement of life in our cities. This is elaborated on in the work of Francesco Bandarin and his colleagues. *Reconnecting the City*[31] stresses that cities are not frozen in time, their historic cores and other areas of heritage value do not exist as separate entities, and that in its entirety, the urban environment cannot be divorced from its greater historical, geographical, topographical, and social context. In this way the adoption of the HUL synthesizes decades of work emphasizing the intangible heritage of the city. Additionally, it opens up greater potential for contemporary conservation policy to improve life in historic urban environments and maintain the sources of vitality, rather than allowing isolation to lead to stagnation. This is summarized powerfully by Patricia O'Donnel's contribution to *Reconnecting* when she states that "the measure of our success in applying the Historic Urban Landscape will be to the degree to which we can clarify objectives for urban heritage to activate the enduring values of urban conservation within the matrix of needs and desires for urban vitality."[32]

2.4.4 Policy and Space Syntax

There is a limited amount of historical research that interacts with the network properties of space.[33] An even smaller portion of that body of research details the application of space syntax as a

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[30] UNESCO, "Recommendation on the Historic Urban Landscape," UNESCO, 2011.
[31] Francesco Bandarin and Ron van Oers, eds., *Reconnecting the City: The Historic Urban Landscape Approach and the Future of Urban Heritage* (John Wiley & Sons Inc, 2015), 5, https://doi.org/10.1002/9781118383940.
[32] Bandarin and Oers, *Reconnecting the City*, 278.
[33] Sam Griffiths, "The Use of Space Syntax in Historical Research: Current Practice and Future Possibilities," in *Proceedings 8th International Space Syntax Symposium*, ed. M. Greene et al. (Environment and Ecology, 2012), 1.

theory or practice for historical research (and fewer still for the purposes of conservation). Of all the English policy documents that this thesis refers to, only one makes specific reference to space syntax methodology. The guidance document "Character and Context: Supplementary Planning Guidance" from 2014, and produced in relation to the 2011 iteration of the London Plan, is intended to sit above borough local plans and provide additional resources for planning with regard to describing "physical, cultural, social, economic, perceptions and experience"[34] of a given area. The toolkit provided in this document recommends, as part of the desk-study portion of any land assessment, the construction of "isovist" maps using space syntax, which would detail the visual permeability of a space. While this is a valid application of space syntax techniques, it differs from the techniques used in this thesis and the body of work on which it builds.

It is apparent then that this area of study is rich with opportunity, especially in a European context. Asian cities, especially those with comparable vibrancy and equally ancient histories are engaging more comprehensively with spatially informed research in pursuit of unifying and enhancing their heritage and planning goals. A number of studies focused on Chinese metropolitan areas are expanding the field of 'heritage syntax.'[35,36] At this time, this thesis can only speculate as to the factors that might explain this trend. For one, China has a deep and continuous relationship with its ancient built environment. Many of the most modern and expanding Chinese cities are built around ancient routes and urban heritages. Furthermore, the infrastructural problems associated with massive populations and rapid urbanization—for context, China has 136 cities with populations of over one million whereas the UK only

[34] Greater London Authority et al., *Character and Context: Supplementary Planning Guidance*, Planning Document (Greater London Authority, 2014), 2.
[35] Han Yue and Xinyan Zhu, "Exploring the Relationship between Urban Vitality and Street Centrality Based on Social Network Review Data in Wuhan, China," *Sustainability* 11, no. 16 (2019): 4356, https://doi.org/10.3390/su11164356.
[36] Jian-Ming Fu et al., "Sustainable Historic Districts: Vitality Analysis and Optimization Based on Space Syntax," *Buildings* (Switzerland) 15 (February 2025): 19, https://doi.org/10.3390/buildings15050657.

has four[37,38]— in addition to extensive tangible and intangible heritage, perhaps necessitates treating heritage studies and spatial network theory as more closely unified disciplines.

2.4.5 Contributing to the Academic Landscape

Local councils have a dense array of statutes and guidance documents from which to build their respective local plans as well as designate and appraise heritage areas. At the same time, this makes it challenging to carry out a comprehensive review of the over 10,000 existing and disparately managed conservation areas.[39] This thesis builds on the insights of existing suburban case studies, and contributes an additional novel case study, in order to examine the relationship between these policies and the implications of space syntax research.

This thesis first reviews the local plans, town center boundary, and conservation area of South Norwood. South Norwood, along with Surbiton and Chipping Barnet, have been examined in meticulous spatial detail by Vaughan et al. and the indicators for their spatial resilience have been described.[40] These town centers are noted for their ability to foster co-presence at various scales of movement due to complex and interconnected street morphology, and movement to and through the town centers is amplified by a wider network of various economic land uses. In short, syntactic analysis of the urban grid demonstrates connectivity both within and beyond the town center, and land use analysis reveals a dynamic range of businesses beyond retail that exist off of the high street itself. People are able to, for example, visit their dental clinic and then go shopping. Others pass through the high street to get to other routes of connection outside of the town, and so on. Most importantly, the journeys overlap in many spaces at many times of day.

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[37] "List of cities in China by population," *Wikipedia*, March 10, 2026, https://en.wikipedia.org/w/index.php?title=List_of_cities_in_China_by_population&oldid=1342728311.
[38] "List of urban areas in the United Kingdom," *Wikipedia*, March 26, 2026, https://en.wikipedia.org/w/index.php?title=List_of_urban_areas_in_the_United_Kingdom&oldid=1345577642.
[39] Historic England, "Conservation Areas," Data.Gov.Uk, January 31, 2025, https://www.data.gov.uk/dataset/f771b8f5-8907-4723-8920-8ed708d22ef6/conservation-areas-service.
[40] Vaughan et al., "An Ecology of the Suburban Hedgerow, or: How High Streets Foster Diversity over Time."

Given that South Norwood only has one conservation area, it presents a good opportunity for an analysis of how the language of its local plan and conservation area appraisal works to recognize the heritage value of this spatial potential, before moving on to a more complex case. Town centers and conservation areas rarely correspond on a one-to-one basis, and so this thesis benefits from an additional and more complex example as shown by Teddington. By presenting this new case, this study demonstrates methodological and conceptual fluency with space syntax research, as well as produces a fresh analysis of the urban vitality of an additional London town.

3. Methods

In each town of study, open source data was used to map conservation areas and town center boundaries, and then produce segment angular analysis. This was completed in QGIS. In the case of South Norwood, the methodology is largely conceptual and serves as a starting point for examining the intersection of syntactic spatial analysis and conservation. The spatial potential and resilience of South Norwood as described by Vaughan et al. was taken as true, and this thesis investigates qualitatively how the single conservation area covering this town engages with, supports, or ignores those findings. This section also demonstrates the complex layering of local policies that dictate suburban planning in London.

This thesis extends the methodology to include an analysis of Teddington. Teddington was selected because it is a peri-urban neighborhood centered on a historic high street, within the two ring roads of Greater London. This aligns with the large-scale survey performed by Vaughan et al. to select their case studies. Additionally, Teddington has the same network classification—district center—and commercial growth potential as South Norwood, as designated by the London Plan. Thus, their relative spatial characteristics and importance to city-wide planning should be loosely comparable. Lastly, Teddington is in a different borough and different geographic context, and has a significantly more fragmented system of conservation, so this thesis can add to existing research with observations from a different local plan and numerous conservation areas.

Data was downloaded from the Space Syntax OpenMapping project, which provided a pre-processed street network for the UK. This data included previously calculated segment angular analysis for *choice* and *integration* at a radius of 2km, normalized for comparative use. This would not be sufficient for the localized needs of this study. Thus, this data was stripped of existing analysis for reprocessing. A 6,000m radius around Teddington was established to account for any edge effects, and then syntactic values were calculated at a radius of 800m and 3,000m. 800m is representative of a 10-minute walk, while 3,000m is the maximum extent of a "local" journey.[41]

For statistical analysis, only the street segments within 3,000m of the Teddington town center were used. From these, the correlation between choice and integration, representative of co-presence potential, was calculated at each radius and further compared to street segment location with regard to town center and conservation area boundaries. For ideological consistency, all conservation areas and town center boundaries within 3,000m were included in the analysis. Thus, the statistical observations include 49 conservation areas and three town centers: Teddington, Twickenham to the north, and Kingston to the southeast.

Taking cues from the previous qualitative analysis of South Norwood's conservation policies, twelve conservation areas in and around Teddington were sampled, analyzed, and scored via a rubric unique to this thesis detailing the extent to which network qualities of space are considered in these

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[41] The process of calculating *choice* and *integration* requires an axial map of a set of streets and a radius for which to calculate. An axial map requires a re-drawing process where each street is reduced to its longest straight-line elements. Streets are then further broken at their intersections to form "segments." From here, the choice and integration for each segment is calculated at the desired radius. Choice measures "how likely an axial line or a street segment is to be passed through on all shortest routes from all spaces to all other spaces in the entire system or within a predetermined distance (radius) from each segment." Integration "is a normalised measure of distance from any space of origin to all others in a system. In general, it calculates how close the origin space is to all other spaces." In practice and for this thesis the process can be thought of as the following: All of the street segments for the UK are downloaded from an open source database. A 6,000m circle is drawn around Teddington. All of the street segments within this circle are included in further calculations. For integration, a street segment is selected, and a path is "walked" to every other street within 800m, and then 3,000m, of that starting place. A value is calculated, and this process repeats for every street in the original 6,000m circle. Streets that require shorter overall trips to reach every other street in the selected radius have higher integration scores. For choice, the process is similar but rather than calculate the shortest distance from a street segment to all other segments, the calculation is related to how often a given street segment is "walked through" on the journey from one segment to all other segments in the desired radius, again 800m followed by 3,000m, from the starting point. For more detailed explanations please refer to https://www.spacesyntax.online/glossary/

documents. Five conservation areas were then explained in individual detail to support this rubric and toolkit.

As there is no expectation that any conservation area appraisal will specifically reference space syntax methodology, scoring was based on the use of language and how it can proxy an understanding of space. Descriptions of conservation areas that include physical locations, relationships to surrounding fabric, street formations and grid densities, movement flows, connectivity, and use were all marked and used to assess scoring. Below is the rubric developed by this study. Each category is scored from 1–3 (1 = absent to poor/few descriptions relating to the category, 2 = adequate and/or semi-frequent descriptions relating to the category, 3 = thorough and/or frequent descriptions relating to the category).

A) **Streetscape significance is related to present-day functions beyond architectural interest**

   The heritage significance of a streetscape is as much related to its current functions as it is to the way that it looks and is experienced. While historical functions are an important aspect of heritage, this thesis suggests that modern day functions are equally important, as they are a continuation of a heritage process that has led to and sustained vitality. The shape and historical development of a set of streets is a critical factor into the types of uses an area can sustain, even if the specific use of a given site has changed over time. Thus, this thesis examines the degree to which descriptions of streets in conservation area appraisals are linked to the way in which they are currently used, and how their morphology has promoted that use. An example of this type of description might link a street's role in connecting to other streets, and how that led to the development of a current retail function. Higher scores will be awarded for more frequent and detailed descriptions of area streets and their present day uses as related to their spatial place within the network.

B) **Recognition of street hierarchies and inter-street relationships**

A result of space syntax analysis is the establishment of a visual hierarchy of spatial relationships. Depending on the type of movement and length of journey, different streets are privileged within a network for different reasons. These distributed privileges are an essential part of the development of urban use and whether or not the areas facilitate co-presence within a network. This thesis looks for language within conservation area appraisals that relates street segments to other streets in the area. This can take the form of noting the ways in which they connect to people, comparing them in terms of relative importance, and any language suggestive of "pervasive centrality"[42] —which would recognize the interlinked networks and localized grid intensification that together create the historic pattern of a townscape, resulting in a system of progressively smaller (or larger) centers. Higher scores will be awarded for more frequent and detailed descriptions of the relative ranking of area streets in relation to other streets in the network.

C) Movement of people in space is considered an aspect of heritage or historical interest

The primary way in which space syntax measures space is by analyzing movement within a network. Given that space syntax posits that the street network is the primary driver of movement dynamics,[43] and that this movement is cyclically related to the creation of space, it is paramount that within their descriptions conservation area appraisals accounts for the ways people move through space. This thesis looks for language that describes directionality—words such as "to," "away," "through,"—and language that describes in any sense the areas in which people gather, use, or avoid. Higher scores will be awarded for more frequent and detailed descriptions of the active movement of people in relation to historical development to the present day.

D) Setting is described beyond vistas and views

[42] Hillier, "Centrality as a Process: Accounting for Attraction Inequalities in Deformed Grids," 15–16; Froy, *Rebuilding Urban Complexity*, 40.
[43] Hillier, "Centrality as a Process: Accounting for Attraction Inequalities in Deformed Grids," 2.

Given that setting is a protected facet of a heritage asset,[44] descriptions of setting are one potential route for more thorough analysis and understanding of technical spatial dynamics. The definition of setting as laid out by the NPPF leaves room for such descriptions, but setting is often only described in terms of vistas and views through space, as well as general aesthetic descriptions of surrounding building typologies or landscapes. This thesis looks for language that suggests an understanding of setting beyond these metrics including townscape, streetscape, and placement within a larger urban network. Higher scores will be awarded for sections within "setting" that define setting in any way other than protected lines of sight.

E) Connectivity and permeability is mentioned as a part of historic setting

The physical movement of people through space is a primary focus of space syntax, but the way in which people are able to move is largely determined by the available connectivity and permeability in space. Connectivity in particular has a direct proxy within space syntax in the form of *integration*, which ranks the most connected and central street segments within a network. Thus, this thesis looks for language within conservation area appraisals that describe streets as connections between spaces, as well as descriptions that describe the number and quality of movement options. This includes, but is not limited to, grid descriptions, the existence of alleys, tunnels, overpasses, bridges, and other indicators that suggest a recognition of centrality, or conversely, isolation. Higher scores will be awarded for more frequent and detailed descriptions of connectivity between places and the methods by which this connection is made and sustained.

F) The built environment and urban form are related to function

Function and form are closely intertwined in spatial dynamics. For example, the long and narrow plots typical in suburban London are often adapted to front-of-house and back-of-house uses, and that front-facing function often changes over time, with respect to, and constraint from, its

[44] *Planning (Listed Buildings and Conservation Areas) Act 1990*, c.9 (UK), s. 66(1).

relationship to the street. A building of this type might be well suited to multiple types of non-residential function and simultaneously limited to a smaller set of retail functions if it fronts the high street. How often and how successfully this use adapts is related both to its form and position.[45] Generally, some built forms support live activity on foreground networks, while others are suited to non-live auxiliary activity (offices, industry, and similar uses) within background networks.[46] Street networks generate patterns of movement over time that affect the way in which the built environment takes shape. This thesis looks for conservation area appraisals that relate the architectural forms of buildings to their historical functions and network placement. Higher scores will be awarded for more frequent and detailed descriptions of how the built environment looks in relation to how people use it, and not just aesthetic character.

G) Boundary placement is described via spatial logic beyond architectural interest

Conservation area boundaries are often defined polygons within more loosely defined settings. These boundaries often align to plot shapes and street geometries serving to encompass specific buildings or landscape features. The strongest boundary delineations will be justified beyond capturing clusters of architecturally or historically interesting features, and define the area by its historic patterns of use and contemporary activities tied to that history. Further, exceptional conservation area boundary logic would account for how areas are interconnected and reliant on one another. For example, the inclusion of streets within the foreground network would include within its boundary the supporting auxiliary streets, even if they were not architecturally relevant. This thesis looks for conservation area appraisals that describe boundary decisions beyond the location of architecturally relevant features. Higher scores will be awarded for more frequent and detailed descriptions of how boundaries were established, and the degree to which these descriptions imply a knowledge of spatial relationships.

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45 Vaughan et al., "An Ecology of the Suburban Hedgerow, or: How High Streets Foster Diversity over Time," 11–15.
46 Froy, *Rebuilding Urban Complexity*, 31, 81, 86, 93.

3.1 Limitations

This study faced a small number of limitations stemming from resource constraints as well as from the novel approach to the qualitative analysis central to a significant portion of this thesis. As there is very little established scholarship directly relating space syntax methodologies to conservation policy review, this thesis is ultimately constrained by a smaller body of work to draw from.

One limitation of the methodology is the inability to qualitatively analyze every conservation area in the statistical analysis due to time constraints. It is worth noting however, that this is itself demonstrative of the different ways that space syntax and boundary-based conservation consider space. Whereas a 3,000m radius can encompass nearly 50 different areas of conservation interest and three towns of varying planning importance, this represents just a fraction of a larger, interconnected network within spatial calculations and observations. The policy side of the equation treats space as inherently fragmented, while space syntax does not.

An additional limitation is the qualitative rubric for scoring conservation area appraisals is unique to this thesis and previously untested. Although it is built on a wealth of spatial scholarship, there is no existing peer-reviewed method for assessing the way qualitative descriptions of heritage can account for essentially quantitative aspects of network theory. Given that both observational and mathematical approaches to assessing urban vitality consider movement, the built environment, streetscape, and use—although approached very differently—the rubric used in this thesis aims to provide a toolkit linking these facets of urbanity across disciplines.

Furthermore, this thesis does not undertake a more thorough review of land use over time. Vaughan et al. provide a compelling case for urban resiliency and vitality through temporal land use analysis,[47] and this thesis is unable to replicate those methods for Teddington. Current land use was sampled and mapped across the focus area and used to provide additional context in some analyses, but results from this sample were not robust enough to provide any strong additional conclusions about spatial

[47] Vaughan et al., "An Ecology of the Suburban Hedgerow, or: How High Streets Foster Diversity over Time," 11–15.

signatures. Further, without the diachronic element, observations of urban resiliency provided by Teddington's spatial characteristics, proxied via continued use, would not be possible. This an interesting potential follow-up to the research presented here.

3.2 Opportunities

This thesis uses a limited set of case studies to describe a workflow for integrating space syntax analysis into conservation area definition, appraisal, and management. Greater London's historical built environment is an essential part of its character and its urban vitality, and it would therefore be inappropriate not to consider what that vitality relies on as part of its heritage management plan. This thesis provides the groundwork for a simplified toolkit for the review of conservation areas with respect to spatial movement dynamics, which in turn could be included in newer versions of conservation area appraisals which are periodically reassessed by local councils.

Although this thesis is focused at the local level and the specific condition of semi-urban London areas, the integration of space syntax and heritage study has applications for planning policy at the local, urban, and national levels.

A compelling continuance of this line of inquiry would be the development of a methodology for a new type of "conservative surgery."[48] This ideological process currently considers the excision of individual buildings. It could instead rely on data and network dynamics to reassess streets and full townscapes to optimize the urban morphology best suited to urban planning goals. This could take the form of the redirection of a street or the reconfiguration of a town block.

Additionally, this thesis enters into a growing dialogue around "heritage syntax," describing ways to incorporate syntactic studies of space into conservation theory and practice. These efforts can go far beyond maintaining vitality in historic town centers and have applications to the areas most sensitive to change. In both the past and present, urban landscapes are often destroyed or threatened with destruction. This can be the result of poor planning, catastrophic ruination, or geo-political turmoil. Rebuilding these

[48] Patrick Geddes and Jaqueline Tyrwhitt, *Patrick Geddes in India* (Lund Humphries, 1947; repr., Lund Humphries, 1947), 40–59, e-book.

areas goes beyond restoring the built fabric and the uses found within them. It must also consider where that area existed within a larger spatial network, and why. How does its loss affect the surrounding areas, and how can it be best reinserted in a way that mirrors its historical-spatial development while maintaining effective links that prevent physical or cultural isolation? Space syntax is a critical tool preventing both the 'Disneyfication' or the ruination of historically vital areas.

4. Study and Results

4.1 South Norwood

The first case study presented in this thesis is South Norwood. To restate, the intention of this thesis is to investigate the overlap between heritage policy, urban complexity, and urban vitality. If heritage is in fact a key piece of urban vitality, then this thesis illuminates the extent to which morphological complexity is considered within heritage policy in the UK at the local level. South Norwood is a case study drawn from the research of Vaughan et al. where it serves as an exemplar of a peri-urban environment of reasonable size and population, centered around a clear economic and historic core. For this thesis, it is also a useful entry point to discuss heritage policy, as it has a single town center boundary enveloped by a single conservation area.

4.1.1 Historical Review of South Norwood

The archaeological record for South Norwood stretches back sporadically to the Roman era, but definitive settlement appears in the mid-eighteenth century around the key intersections of Selhurst Road and South Norwood Hill, and Whitehorse Lane and South Norwood Hill. Portland Road was and continues to be South Norwood's main passage on its north-south axis, connecting it to London. South Norwood continued to grow radially from these key intersections. A brief incursion of the Croydon Canal shaped approximately thirty years of development in the early 19th century, and its curved course can still be seen in the roadways and crescent development of Regina Road, Lincoln Road, Eldon Park, and Albert

Road. Further and more dramatic growth, as well as a degree of spatial determinism, came with the arrival of the railway at Norwood Junction, bringing a large influx of people and economic activity while also bisecting South Norwood. Both Portland Road and High Street had their turn as commercial centers based on the moving location of Norwood Junction in the latter half of the nineteenth century. Today, much of South Norwood is well preserved as a largely Victorian town characterised by an active commercial core and residential side streets of historic character and form (figure 3a-d).[49] The conservation area appraisal and management plan of South Norwood cites this architectural character as a primary factor in its heritage contribution.[50]

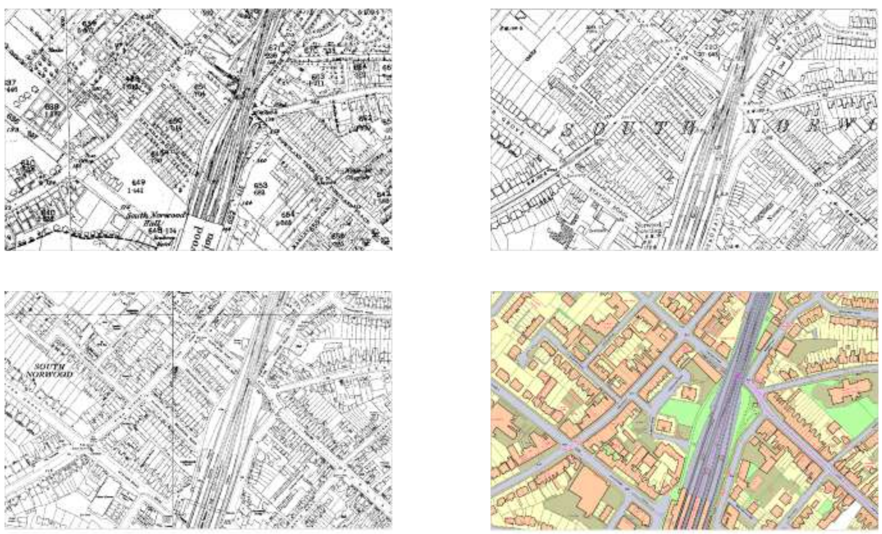

Figure 3a-d:
Ordnance Survey maps for South Norwood town center.
c.1880, 1910, 1960 and 2013 - left to right, top to bottom. Scale 1:1500.

© Crown Copyright/ database right 2013. An Ordnance Survey/EDINA supplied service.

Source: Vaughan et al., "Beyond the Suburban High Street Cliché - A Study of Adaptation to Change in London's Street Network: 1880-2013," 226.

4.1.2 The Vitality of South Norwood as a Product of Space

The fact that South Norwood, despite tumultuous economic conditions, remains an active town center with a lively high street and is the heart of its respective community is itself evidence of its resiliency. It is this community longevity and adaptability that Vaughan et al. set out to understand

[49] Vaughan et al., "Beyond the Suburban High Street Cliché - A Study of Adaptation to Change in London's Street Network: 1880-2013," 226.
[50] Place Services et al., "South Norwood Conservation Area Appraisal and Management Plan," Supplementary Planning Document, Croydon Council, 2022, 9.

through their spatial research. The markers for this spatial potential can be summarized as follows: multi-scale co-presence, a range of adaptable building forms across numerous small plots, and a diversity of mixed uses covering a geographic range outside of the boundaries of the retail center.

Multi-scale co-presence suggests the capacity for people to exist in space simultaneously when on journeys of different length, at multiple places in an urban network. When Vaughan examined the correlation between to-movement and through-movement at radii of 800m and 3,000m, she found a correlation at both scales, although far less so at 3,000m. In other words, the street morphology facilitates sufficient movement scale and type for economic activity to take advantage of. As for adaptable building morphologies, a land use assessment showed the persistence of use—whether economic, social, or communal—in many buildings. Storefronts are often converted to other storefronts, and back-of-lot spaces can be used for various industrial, storage, or "non-live" activities, for example. On the other hand, Vaughan points out the comparative weakness of co-presence at larger scales, especially compared to her other case studies, and the subsequent observation of lesser economic success in South Norwood, particularly in areas that facilitate large-scale movement, such as those around the train station

The study describes the network of non-retail activity that exists at the fringes and beyond the traditional core of South Norwood, including offices, services, and community uses. Vaughan concludes that this wider range of attractors is both seeded by the spatial morphology of South Norwood and further directs movement to and through the high street, as people are able to complete multiple errands in a single trip. As opposed to the "live center" described by Hillier, Vaughan suggests a larger and dynamic "active center" exists in London suburban contexts that is essential to the overall vitality of the town centers.[51] This mix of spatial signifiers is what this thesis considers "urban complexity" and what conservation policy is being benchmarked against in this study.

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[51] Vaughan et al., "Beyond the Suburban High Street Cliché - A Study of Adaptation to Change in London's Street Network: 1880-2013," 225, 238.

4.1.3 Conservation in South Norwood and Space Syntax

The management of the conservation of the built environment in South Norwood is directed at the borough level, itself empowered and guided by the national frameworks and policies discussed previously. What follows is the review of these policy frameworks and an assessment of where the principles of space syntax analysis could lead to improvements.

South Norwood is in the borough of Croydon and is designated a "district center" in accordance with the London Plan[52] hierarchy, which seeks to appropriately plan the growth of the network of town centers across Greater London based on their size and economic potential. Based on its size, South Norwood's planning is focused mainly on satisfying economic, sustainability, and housing goals—i.e. vitality—at a localized level.

The Croydon local plan, as it pertains to South Norwood, is heavily focused on the formal town center boundary, and designation of land use within and outside of this area, specifically keeping primary town center uses (retail, restaurants, cafes, and major businesses) within this boundary so as to concentrate the economic life of the area into the area deemed most accessible to the greatest number of people.[53] Note that this accessibility is mainly grounded in historical use as well as notions of public transportation. The notion of polygonal boundaries is at odds with the findings and principles of space syntax, and Vaughan et al. make a convincing argument that "primary town center uses" is a retail-centric definition that undercuts the importance of the surrounding land uses.[54] Additionally, boundary-based definitions of the town center, at least in the case of South Norwood, do not account for the historical movement patterns and current grid conditions that have led to the emergence and sustainability of this area.

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[52] Greater London Authority, "The London Plan: The Spatial Development Strategy for Greater London," Greater London Authority, 2021, 479, 482.
[53] Special Planning Service London Borough of Croydon, "Croydon Local Plan 2018 (Revised 2021)," London Borough of Croydon, 2018, Policy DM8.
[54] Vaughan et al., "Beyond the Suburban High Street Cliché - A Study of Adaptation to Change in London's Street Network: 1880-2013," 238–39.

South Norwood also has two "neighborhood centers," a smaller hierarchical designation meant to service the immediate local population with goods and services. These neighborhood centers are identified as being "well connected" and while there is an overall aim of reducing vehicle use in these areas, their connectivity to the town center is not identified, nor are the historical spatial relationships that have led to the emergence of these hyperlocalities or their spatial dependencies on the greater town center, both of which could be revealed by space syntax analysis.

South Norwood's specific heritage assets are defined in the Local Plan and managed by the conservation area appraisal and management plan for South Norwood. The town contains only one conservation area, which itself fully envelopes the formal town center. Conservation areas have the most restrictive planning rules and cannot be targets of any densification schemes. In fact, all development must "preserve and enhance heritage assets as their settings."[55] Insofar as the historic built environment is a central aspect of placemaking within the Local Plan, there is a recognition that both the areas in which the community is most likely to interact and the means by which they travel between these areas must be distinct and maintain their local character. However, these places are presumed to be town centers and main roads without further investigation into evidence supporting these claims. A "key strategic roads" hierarchy framework is identified as a future output of the Local Plan,[56] but it does not exist either as a completed framework or as a set of tools for developing such a categorization. Space syntax could serve here as an essential toolkit for providing data-backed evidence as to the local significance and greater significance of connective routes between town centers.

Turning to the conservation area appraisal itself, this thesis looked for evidence that the appraisal described the historical significance of movement networks, land use and land use relationships, the streetscape beyond the architectural character of its buildings or its experiential qualities, and how the history and heritage of South Norwood are tied to movement in and through the larger setting of the heritage area.

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[55] Special Planning Service London Borough of Croydon, "Croydon Local Plan 2018 (Revised 2021)," 96.
[56] Special Planning Service London Borough of Croydon, "Croydon Local Plan 2018 (Revised 2021)," 96.

South Norwood was first designated a heritage area in 1992 and was subsequently expanded in 2007 to include additional residential side streets and cross the railway line.[57] The 2022 expansion added Holmesdale Road, Lawrence Road, Cargreen Road, Portland Road, Albert Road, Warminster Road, and Lancaster Road. In each of these individual considerations, only the significance of the architectural character and its preservation is considered, and not the collective roles of these streets in sustaining the presence and activity of people. The epicenter of vital activity in South Norwood is at the junction of Portland Road and High Street, and yet the 2022 extension is justified only by the inclusion of the locally listed Mission Hall and the subsuming of a row of buildings listed as a local heritage area. The description of South Norwood's special character does reference the intact layout of the Victorian buildings and streets, but only insofar as they are evidence of the past and not as they function to the vitality of the present. Further definitions of the setting of the South Norwood conservation are a non-exhaustive list of surrounding streets that have also retained their architectural character.

More nuanced spatial considerations are included in the descriptions of "townscape character" where the appraisal characterisations protect the tight urban grain, narrow plots, the constitution of streets by building frontages, and the mix of commercial and residential land uses. This section more or less guarantees the survival of the urban grid and indicates the "arterial" nature of High Street and Selhurst Road, which is suggestive of their importance as movement-based heritage, as supported by space syntax analysis.

Beyond these illustrations, the conservation area assessment does not address the function of South Norwood's streetscape as an interior network of co-presence. It also overlooks the spatial relationships (historical or otherwise) to the rest of the urban fabric and extended class of non-residential uses. This raises the question as to how space syntax analysis could provide a deeper understanding of this heritage asset and what maintains its vitality.

[57] Place Services et al., "South Norwood Conservation Area Appraisal and Management Plan," 4.

4.2 Teddington

To put this question to the test, this thesis presents an additional case study in Teddington. The previous chapter focused on the existing indicators of spatial potential in South Norwood as identified by previous research and examined how this urban complexity was or was not covered by the regional spatial management frameworks. In this chapter, Teddington serves as a clean slate to examine both the extent to which spatial potential exists and can be identified by space syntax methods, and whether it is addressed in the conservation management frameworks. Teddington additionally provides an interesting case in that it is in a different borough than South Norwood and other suburban centers examined in previous research. Furthermore, the conservation areas around the town center are exceptionally fragmented (figure 4).

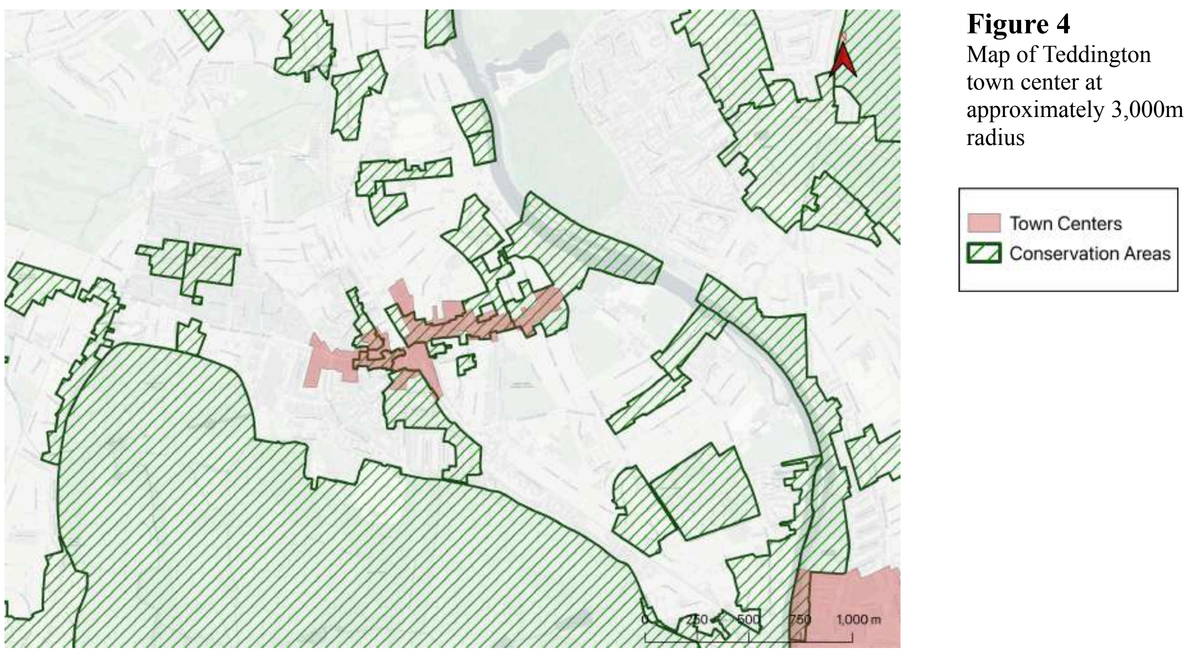

Figure 4
Map of Teddington town center at approximately 3,000m radius

Source: Author's own diagram. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

Spatially this is an interesting dynamic and also adds a new layer of complexity from a policy perspective. Each of the conservation areas observed in this thesis is individually appraised and contains different management strategies, even though they together fall under the jurisdiction of the Royal Borough of Richmond on Thames.

Within the entirety of the 3,000m study area, there are 49 conservation areas and three formal town centers. The qualitative analysis presented in this thesis looks at the 12 conservation areas closest to Teddington Town Center, excluding Bushy Park due to its size and intention of conserving green space, which is out of the scope of this thesis. The statistical analysis includes all 49 conservation areas and considers whether or not a given street segment is wholly contained, partially contained, or outside of any conservation area or town center within the 3,000m study area.

4.2.1 Teddington at Large

Teddington is one of five town centers in the Borough of Richmond on Thames. While the formal town center boundary includes only High Street and some of Broad Street, the town itself roughly extends between Hampton Hill to the west, the River Thames to the east, and Bushy Park and Hampton Court to the south. Much of Teddington's history is defined by its relationship to the river.

Teddington remained largely isolated and affluent throughout its early history, but additions of the Teddington Lock and subsequent bridge, as well as rail access, eventually led to a population boom, and the town retains much of its Victorian architectural character.

The Local Plan for Richmond designates Teddington as a district center and recognizes its potential value for economic growth both for servicing current residents and visitors. Teddington is also seen as a local gateway to the neighboring green spaces and Hampton Court.

A theme throughout the local plan is the accomplishment of the 20-minute city while navigating the wide reach of protected assets in the borough that would challenge development strategies. This emphasis on "living locally"[58] is markedly different from the Local Plan of Croydon and places a specific emphasis on movement and connectivity. Additionally, the Local Plan contains specific language about considering high streets beyond their retail potential and investigating and protecting the mix of uses

[58] Richmond Upon Thames Local Plan: "The Best for Our Borough" (2024 to 2039) (2025), 18.

sustained throughout the historic development of town centers that support the central high street vitality. This language mirrors the conclusions of Vaughan et al. and makes Teddington and the Borough of Richmond on Thames good candidates for space syntax analysis and what it can reveal with regard to planning goals.

Insofar as the Local Plan plans for the adaptability and "long term stability"[59] of town centers and high streets, in addition to connectivity between town centers, this raises the question of what space syntax analysis reveals about the spatial qualities of Teddington that promote these goals.

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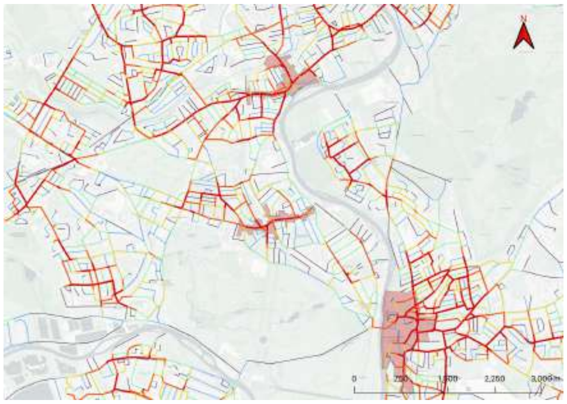
[59] Richmond Upon Thames Local Plan: "The Best for Our Borough" (2024 to 2039), 252.

4.2.1.1 Space Syntax Analysis of Teddington Townscape

Choice at 800m



Choice at 3,000m

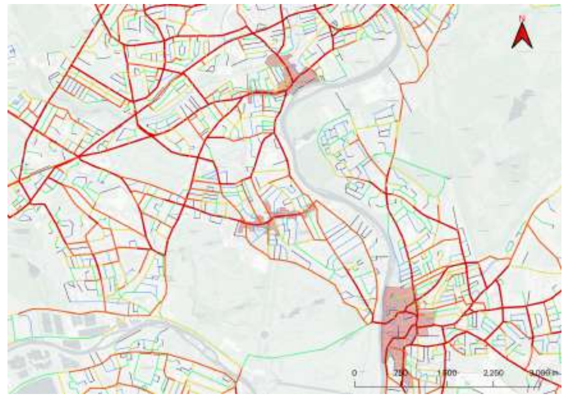
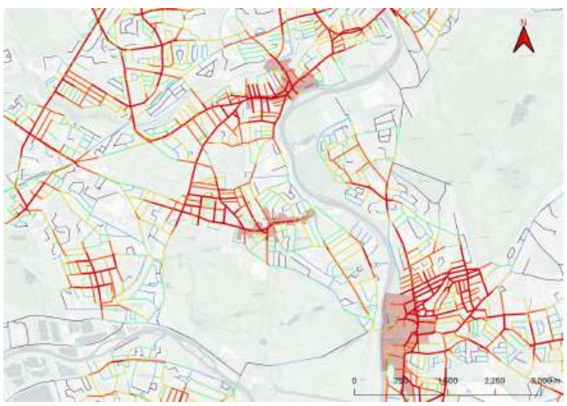
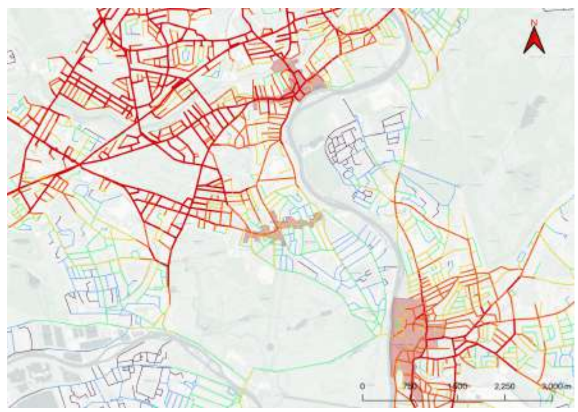


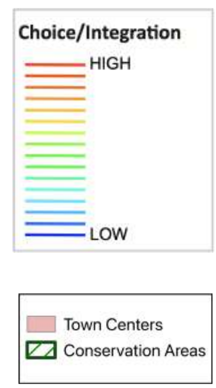
Figure 5a-f
Space syntax model of choice, integration, and combination, at radius 800m (left) and 3,000m (right) in the Teddington study area in sequential order
(top to bottom).

Integration at 800m

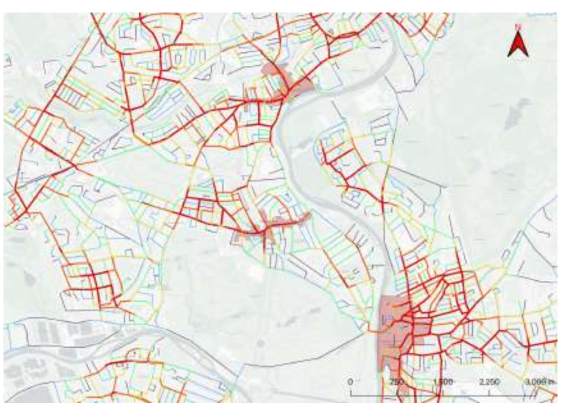


Integration at 3,000m





Combination of choice and integration at 800m



Combination of choice and integration at 3,000m

Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

Figures 5b-5e show segment angular analysis for choice and integration at radii of 800m and 3,000m respectively. Figures 5f-5g show the combination of these syntactic measures. The measures of choice, or through-movement, at both scales show a strong north-south axis of movement in the Teddington area. This is likely because of the limited number of river crossings to the east and the town centers of Twickenham and Kingston to the north and south. The data used in this study accounts for only one significant river crossing, connecting the southernmost tip of Teddington to Kingston, which further stretches the vertical axis of movement through the town. At smaller scales of movement, Kingston Road and Broom Road emerge as significant routes of local through movement, whereas at the large scales emphasis is also placed on Sandy Lane, the westernmost of the three vertical access routes to Kingston.

The grid intensifies to the north of the town center, where more movement potential and spatial complexity can be seen. The southern half is far less connected, and this relative isolation is most stark when comparing integration at 800m and 3,000m. The lack of high-value “red” lines at the larger scale suggests limited connectivity and “closeness” in this area, and its spatial potential is thus reduced. This limited integration at large scales extends concerningly to the high street itself, indicating a more limited capacity for adaptability and resiliency in the area that needs it most.

Observationally, there is a suggestion that the relationship between Teddington and Twickenham is closer than between Teddington and Kingston, which logically follows from the fact that they are in separate boroughs. However, one stated overall goal of the London Plan, which sits above the respective local plans of these boroughs, is improved town center connectivity, and this spatial analysis already shows the ways in which those spatial relationships may be limited or strained. Additionally, at the borough level vitality is a stated goal. Given the relationship between connectivity and vitality, it would be wise for councils to consider integration amendments with regard to facilitating journeys at a larger scale. There are four conservation areas in this region of Teddington, and should any of them overly restrict the borough’s ability to act on spatial interventions that would conserve urban vitality, they may in reality be working counterintuitively. Thus, not only can space syntax thinking expand the scope of conservation areas, but it may also help to limit them or contextualize their reach.

Co-presence in Teddington can be proxied via the combination of angular choice and integration, where high-value “red” lines represent areas where people are more likely to be co-located in space at a given time. Unsurprisingly, High Street demonstrates a high level of co-presence, which is directly related to its spatial qualities. However, co-presence can also be seen immediately to the northwest of the high street, and it is important to consider what role this plays in supporting the high street itself. The spatial characteristics of this area are likely related to moving people to and through the high street and sustaining its vitality.

The spatial analysis can thus add a quantitative and observational layer characterizing the town center's urban complexity that the architects of the Local Plan can use to better inform policies in pursuit of their goals. As noted previously, one such goal is the protection and maintenance of town center vitality, and this thesis posits that conservation areas are one such protective framework. The degree to which Teddington’s conservation area appraisals account for the spatial complexity described is explored in the next section.

4.2.2 Aggregate Results: Qualitative and Quantitative

The qualitative analysis of the conservation area appraisals and management plans consisted of routine observations of the language contained in these documents and a final scoring across seven categories relevant to spatial research aims, as described in the chapter on methods.

4.2.2.1 Qualitative Assessment Results

Table 1: Conservation Area Qualitative Assessment Results								
Conservation Area	Streetscape significance related to present-day functions beyond architectural interest	Recognition of street hierarchies and inter-street relationships	Movement of people in space is considered an aspect of heritage or historical interest	Setting described beyond vistas and views	Connectivity and permeability mentioned as a part of historic setting	Built environment and urban form related to function	Boundary logic described via spatial logic beyond architectural interest	Average
Fieldend	2	3	2	3	3	3	1	2.43
Royal Road	1	1	1	1	1	2	1	1.14
Mays Road	2	2	2	2	2	2	1	1.86
The Grove	1	1	1	1	2	2	1	1.29
Teddington Lock	2	2	1	2	3	3	2	2.14
High Street Teddington	2	2	1	2	1	3	2	1.86
Blackmore's Grove	1	1	1	1	1	1	1	1.00
Church Road	2	2	1	1	1	1	1	1.29
Broad Street	2	1	1	1	1	2	1	1.29
Park Road (Teddington)	1	2	2	1	2	2	1	1.57
King Edwards Grove	1	1	1	1	1	2	1	1.14
Wick Road	2	1	1	2	2	2	1	1.57
Average	1.58	1.58	1.25	1.5	1.67	2.08	1.17	

Among the 12 conservation areas analyzed, the relationship between built form and function emerged as the category with the highest average score. This thesis conjectures that this may be because conservation area appraisals are usually focused on describing buildings. In describing the form of the built environment, the function of the building is more likely to be mentioned as well. While an explicit connection between form and use is not always drawn, this trend likely resulted in higher average scores.

The lowest scoring categories were in describing boundary logic and descriptions of the movement of people. These are arguably the most critical categories within the greater context of this study, as this thesis argues that space syntax, which is based on movement dynamics, is a potential tool to reconsider conservation area boundary placement based on a more coherent understanding of the space in question. It is unsurprising that these two assessment categories scored lowest, given that conservation area logic is often limited to the static environment and boundary definitions follow suit. Setting fixed boundaries can even be considered at odds with greater network thinking.

Among individual conservation areas, Fieldend had the highest average scores. Perhaps because of its location and relatively small size, it is of greater necessity to define it with regard to space rather than character. A specific note on Fieldend is included in the following chapter.

Teddington Lock also scored above average. Perhaps because the lock itself is a historical artifact of movement rather than a dwelling or non-residential building, it is easier to define it with regard to movement and space overall. A section dedicated to Teddington Lock is included in the following chapter.

Blackmore's Grove was the lowest scoring single conservation area appraisal in the study. The appraisal made no effort to define any sense of spatial dynamics, placement, or connectivity, and was defined solely by the architectural character of the buildings included. The location of this set of buildings is subsequently the only factor in determining the boundary of the Blackmore's Grove Conservation Area.

A potential emergent pattern is that conservation areas based on groves, estates, and closed developments are less likely to use spatial language in their appraisals as opposed to areas based on streets and roads. This may appear obvious based on the differing nature of these two features of the built environment, but this study provides a small set of data in support of this observation.

4.2.2.2 Aggregate Quantitative Results

The quantitative analysis first calculated the correlation between *choice* and *integration* as a proxy for co-presence, as was previously done by Vaughan et al.[60] That study presumed that where this correlation was high, co-presence and urban complexity were also high. Where spatial morphology resulted in this measure being highly correlated at multiple scales, 800m and 3,000m, it was indicative of overall resiliency and vitality for that town center.

[60] Vaughan et al., "An Ecology of the Suburban Hedgerow, or: How High Streets Foster Diversity over Time," 4–11.

This study utilizes the same measure, but instead of calculating it for all roads in the network, streets are grouped by their position based on containment within conservation areas and formal town center boundaries. The first quantitative analysis compared both movement scales grouped by whether streets are wholly contained, partially contained, or not included in any conservation area within the 3,000m radius around Teddington.

The second analysis again compared both movement scales, grouping streets by cross-referenced containment within conservation areas *and* town center boundaries. The same 3,000m radius was used, necessitating the inclusion of the town centers of Twickenham and Kingston given their relative proximity to Teddington. Streets were likewise divided into wholly contained, partially contained, and not contained categories for a total of nine possible categories at each scale of movement.

Correlation between Choice and Integration at 800m and 3,000m by Conservation Area Containment Status

Table 2.1: Correlation between Choice and Integration by Conservation Area Containment Status		
Correlation of Choice and Integration at:	800m	3,000m
Inside Conservation Area	**0.8548**	0.6756
Partially in Conservation Area	**0.7949**	0.5205
Outside Conservation Area	0.7531	0.5999

When comparing only whether a street is inside or outside of a given conservation area, it is apparent that co-presence is slightly higher for streets in conservation areas than for those not included at all. This follows the logic that conservation areas in and around town centers are likely to include roads of higher importance for live activity. However, it is worth noting that correlation between choice and integration for streets outside of conservation areas is also relatively high. This suggests that conservation areas do not exclusively *target* areas of high co-presence, despite the importance of this measure for urban vitality. At the larger movement scale, the correlation in all three categories breaks down markedly, with co-presence only loosely present regardless of which group is being assessed. Should a number of these conservation areas be centered on areas of urban life, the lack of co-presence at large scales of movement is suggestive of lower levels of urban resiliency.[61]

Correlation Between Choice and Integration at Different Movement Scales by Conservation Area and Town Center Containment Status

Table 2.2: Correlation Between Choice and Integration at 800m by Conservation Area and Town Center Containment Status			
Correlation of Choice and Integration at 800m	Inside Conservation Area	Partially in Conservation Area	Outside Conservation Area
Inside Town Center	**0.8663**	0.7415	0.6793
Partially in Town Center	**0.8825**	**0.9273**	0.7028
Outside Town Center	**0.8711**	0.7750	0.7656

Table 2.3: Correlation Between Choice and Integration at 3000m by Conservation Area and Town Center Containment Status			
Correlation of Choice and Integration at 3,000m	Inside Conservation Area	Partially in Conservation Area	Outside Conservation Area
Inside Town Center	0.7279	0.5174	0.6009
Partially in Town Center Partially	0.6675	0.7454	0.5369
Outside Town Center	0.6407	0.5109	0.6549

When comparing both containment within a conservation area and town center boundary, there is again greater correlation between choice and integration at smaller scales of movement. At local movement scales, the relationship between co-presence and conservation area appears stronger than town center containment, with the correlation higher for streets within conservation areas regardless of whether they were in the town center or not. It is equally notable that location within a town center is not necessarily related to co-presence, which seems counter-intuitive.

An additional observation is that, at the smaller movement scale, the highest correlation emerged within the group of streets partially contained within both a conservation area and town center boundary. This is possibly representative of the fringe streets that radiate from the high street, which itself would typically be the center of focus for both conservation area appraisals and town center boundaries. Given that this network of streets is critically important for facilitating connection and movement to and through

[61] Vaughan et al., "Beyond the Suburban High Street Cliché - A Study of Adaptation to Change in London's Street Network: 1880-2013," 238.

the town center, and that there is a measurable amount of spatial potential here, it could suggest that these boundaries are incomplete. Local councils may want to reassess these designations to fully recognize the spatial morphologies and potentials that their towns are predicated on.

At the larger movement scale, the correlation drops significantly in all categories, suggesting that the towns in the study are failing to produce co-presence at this measurement level. In this second analysis, being partially included in a conservation area appears negatively related to co-presence. Again, this could be indicative of incomplete boundary appraisals or a result of the significant fragmentation of these areas. Perhaps conservation areas are failing to integrate with each other and with neighboring town centers.

There is an overall suggestion from the data that conservation areas and the town centers in the study are currently only serving a hyper-local function. This could present a threat to future vitality, as areas dependent on the activity generated by people passing through this space are not capturing visitors on longer-scale journeys. This could be for a number of reasons. For example, people may be moving from Twickenham while bypassing or avoiding Teddington; or it could be related to the lack of river crossings in the Teddington area forcing diversions or alternative routes of travel. This is especially concerning given Teddington's status within the London and Local Plan as a "district center,"[62] placing it higher in the administrative hierarchy with regard to the geographic reach of its functions and urban value.

[62] Greater London Authority, "The London Plan: The Spatial Development Strategy for Greater London," 482.

4.2.3 Conservation Area Deep Dives

4.2.3.1 High Street Teddington[63]

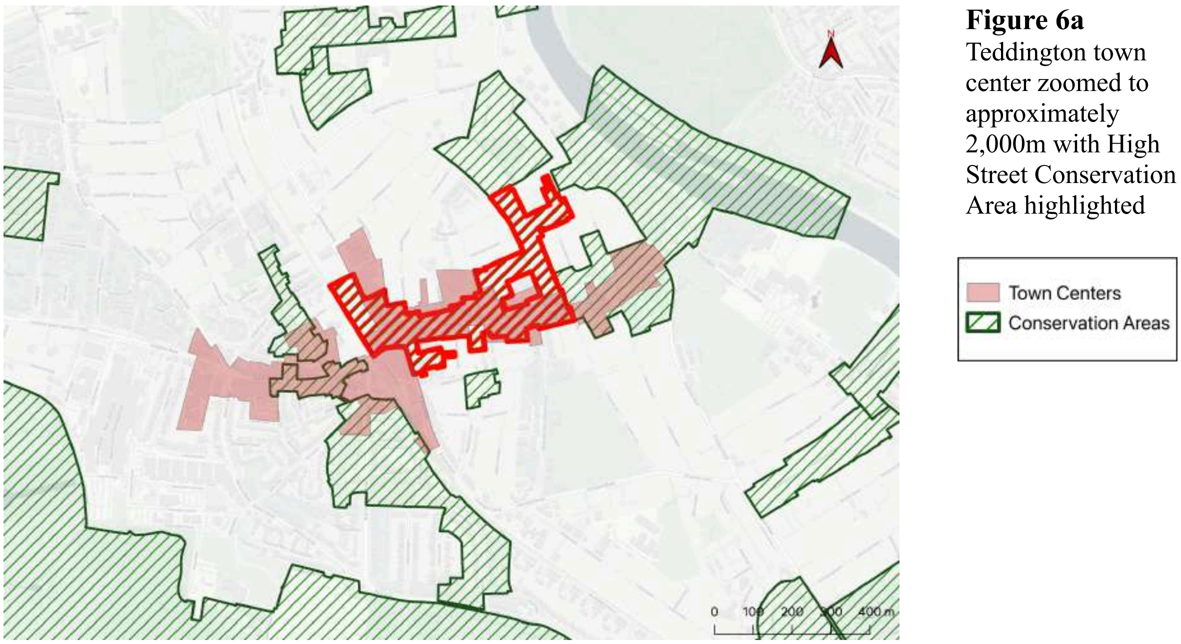
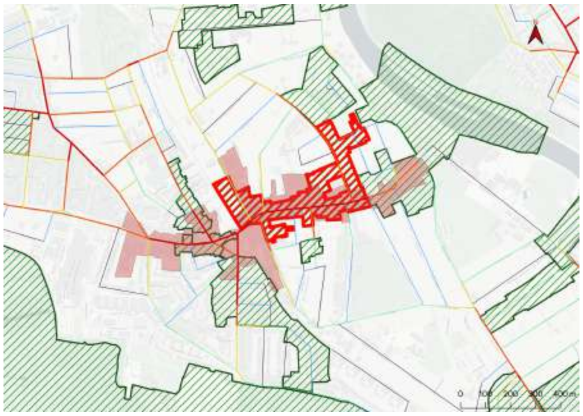


Figure 6a
Teddington town center zoomed to approximately 2,000m with High Street Conservation Area highlighted

Source: Author's own diagram. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

[63] Royal Borough of Richmond on Thames, *High Street Teddington: Conservation Area Appraisal Conservation Area No. 37* (Environment, Sustainability, Culture and Sports Committee, 2023).

Choice at 800m



Choice at 3,000m

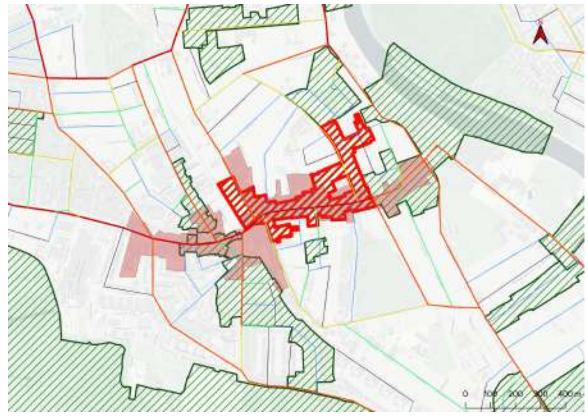
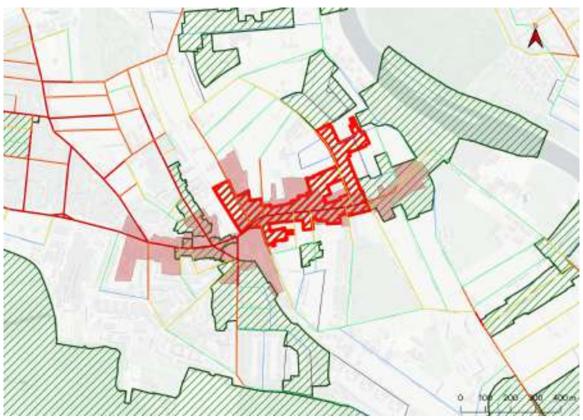
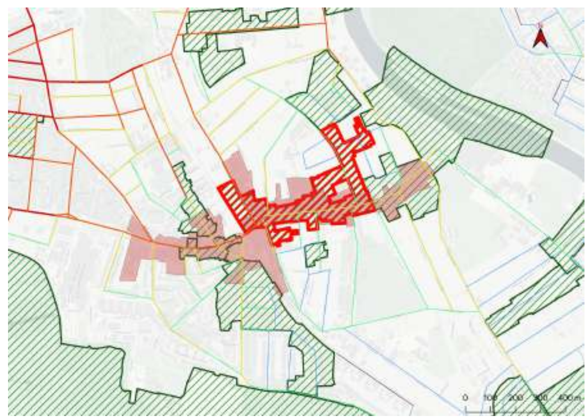


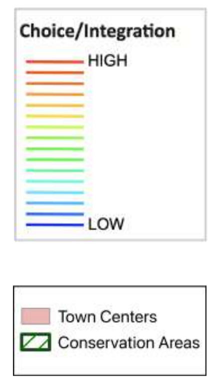
Figure 6b-g
Space syntax model of choice, integration, and combination, at radii 800m (left) and 3,000m (right) in the High Street Conservation Area study area in sequential order (top to bottom).

Integration at 800m

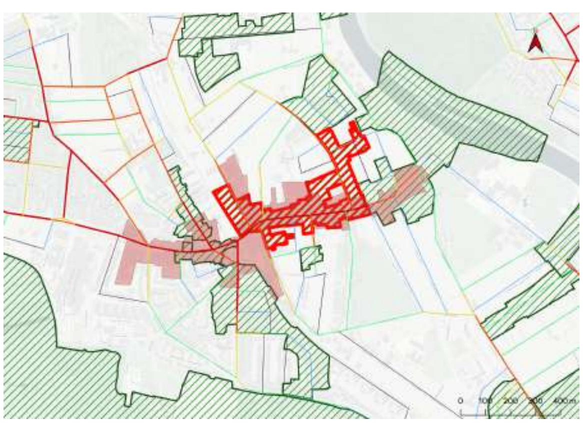


Integration at 3,000m

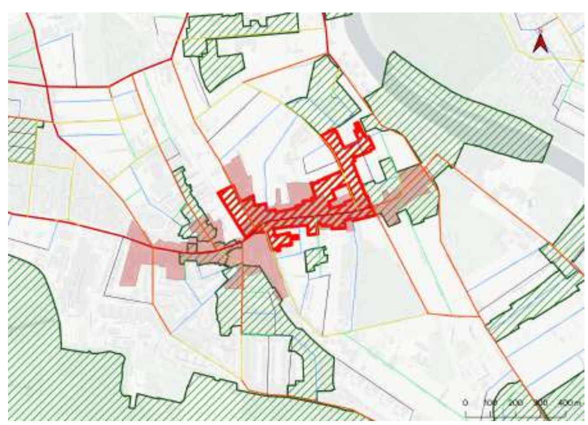




Combination of choice and integration at 800m



Combination of choice and integration at 3,000m



Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

The High Street Conservation Area in Teddington is a multi-faceted polygon covering most of the High Street except where it abuts the neighboring Teddington Lock Conservation Area. It is oriented mostly west to east and covers some of Waldegrave Road and Cambridge Road respectively, but not their full lengths. Also notable is a gap in the center of the boundary that leaves several lots uncovered. The auxiliary streets to High Street are also partially covered, and more heavily to the north than south.

Segment angular analysis of choice and integration at 800m demonstrates the importance of the high street as both a destination and connective tissue (figure 6b and 6d). There is therefore some alignment of the conservation area and basic network importance. However, when examining the combination of choice and integration at 800m (figure 6f), a proxy for co-presence, much of the length of Waldegrave Road to the west demonstrates this spatial potential yet remains uncovered by the conservation area boundary.

Although the focus of the High Street Conservation Area is to protect the high street, the lack of consideration for the north-south throughways that provide activity is apparent, and this is even more pronounced at the 3,000m scale (figure 6g).

The high street is not as well integrated at larger scales than would be expected (figure 6e), suggesting that its "closeness" relative to the broader network is more limited. Should anything disrupt the north-to-south passages that lead to the high street, its resilience and vitality could be severely hampered. This is an example of where space syntax analysis could improve the understanding of this conservation area, suggesting that an aspect of its setting are the spatial relationships to the streets that feed its activity, especially at larger scales. Space syntax mapping shows a clear network relationship between Twickenham to the north and Teddington, but conservation area appraisals rarely address dependencies at this scale.

Other areas of weakness in the area appraisal include remarks that Marvan Court lacks heritage importance, due to a lack of architectural interest, and yet the spatial analysis shows that this is a critical junction between High Street, Park Road, Station Road, Broad Street, and Waldegrave Road. This research found no mention of connectivity or the movement capacity in the conservation area, and boundary logic is defined primarily by architectural interest.

The conservation area appraisal does substantially address the relationships between building forms and urban grain, and successfully relates current use to spatial morphologies, noting, for example, that the form of the high street has historically facilitated front-facing purpose-built shops.

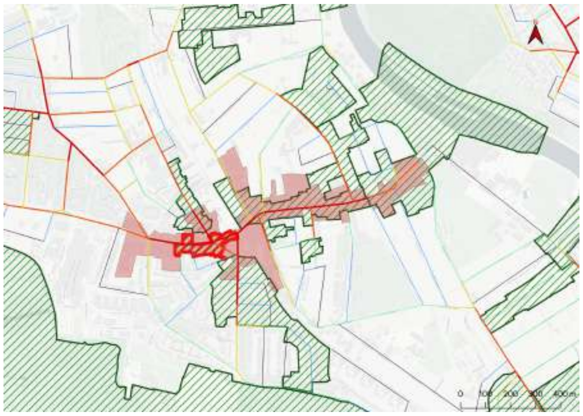
4.2.3.2 Broad Street[64]

Figure 7a
Teddington town center zoomed to approximately 2,000m with Broad Street Conservation Area highlighted

Source: Author's own diagram. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

[64] Urban Design and Conservation et al., *Broad Street: Conservation Area No. 84*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2021).

b) Choice at 800m



c) Choice at 3,000m

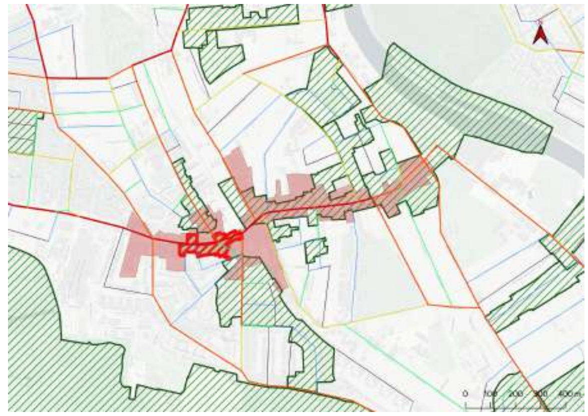
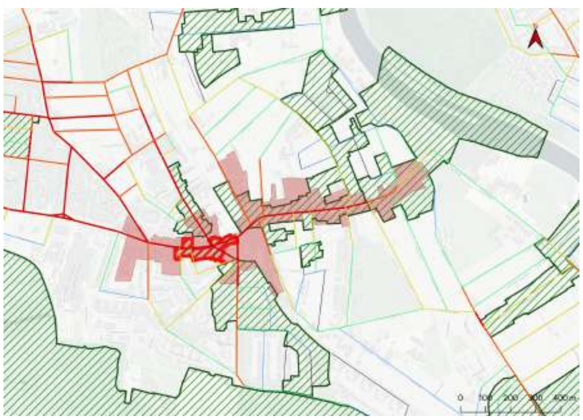
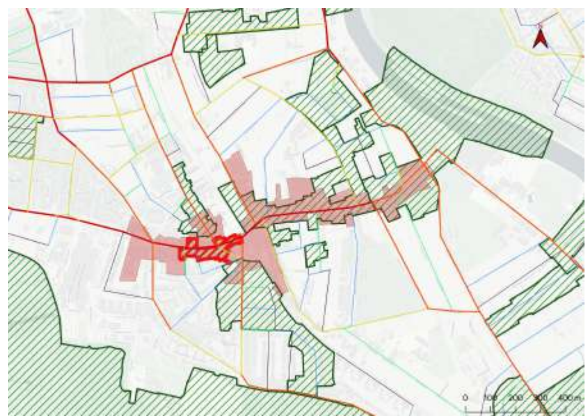


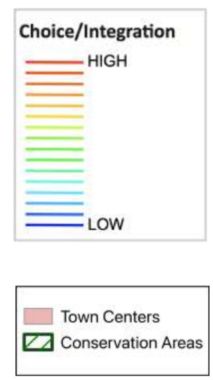
Figure 7b-g
Space syntax model of choice, integration, and combination, at radii 800m (left) and 3,000m (right) in the Broad Street Conservation Area study area in sequential order (top to bottom).

d) Integration at 800m



e) Integration at 3,000m





f) Combination of choice and integration at 800m

g) Combination of choice and integration at 3,000m

Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

Broad Street is the second most commercially active street in Teddington and more integrated at multiple scales than the neighboring High Street, suggesting a high degree of spatial potential and adaptability. Despite this, the conservation area designating Broad Street only covers about half of its geography. The western end is not included in any conservation area despite its relative network importance and the mix of uses seen here (see figure 7h). Although this could be addressed as part of the setting of Broad Street, the western portion is not designated in this way within the area appraisal. At both the small and larger scales, the syntactic analysis shows that Stanley Road presents itself as an important through-route (high choice) likely directing people to Broad Street. The same can be said of Queen's Road at the larger movement scale. These streets are therefore important to the functionality of Broad Street and yet are unconsidered in the area appraisal (figures 7b-g).

Broad Street demonstrates a general capacity for co-presence important for its ongoing resilience, although it is worth noting that this is observably lesser at the larger movement scale. There are also notably fewer non-residential uses to the south of Broad Street than to the north, where the area is less integrated and co-presence is generally lower.

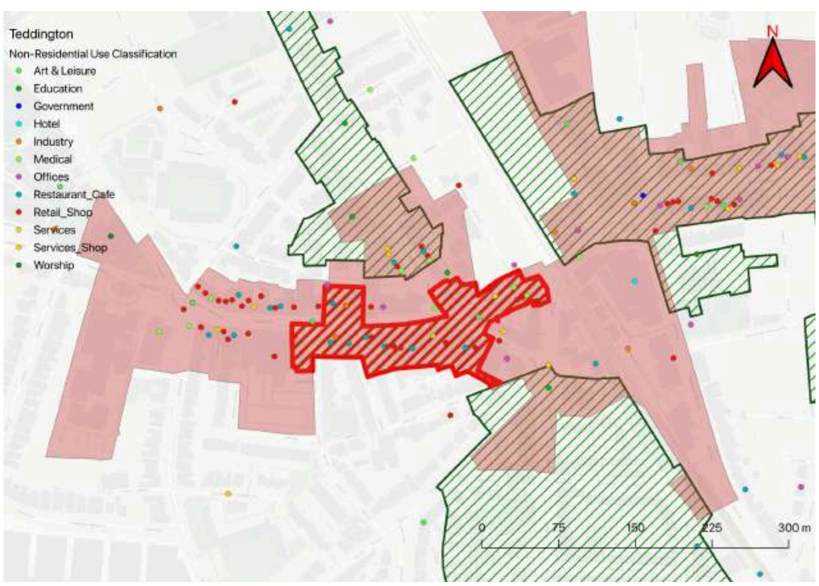
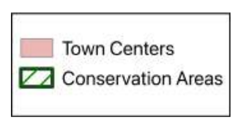


Figure 7h
Sample of non-residential land use distribution in Teddington with Broad Street Conservation Area highlighted.



Source: Author's own diagram. Land use data extracted via Google Maps Scraper (see https://console.apify.com/actors/nwua9Gu5YrADL7ZDj/information/latest/readme, by Compass, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

The boundary of the conservation area is defined by experiential and architectural interest along the route covered and slightly to the north, although it notably cuts short of a single plot on Church Road, whose conservation is then covered by a separate area beyond that plot. This is another example of fragmentation over network consideration.

The area appraisal for Broad Street recognizes the Victorian and Edwardian character of the street and its historical function as a route into Teddington, which is a notable consideration of movement dynamics not included in every area appraisal. There is equally a thorough accounting of building forms and their relationships to the urban grid, but this is ultimately the limit of spatial thinking in the appraisal.

The spatial analysis shows that Broad Street occupies an extremely privileged position in Teddington, facilitated by its relationships to neighboring street segments. Conserving urban vitality in this area should account for this. Broad Street would be at risk if something were to significantly alter the spatial morphology of Stanley Road or Hampton Road for example, which would in turn affect the High Street itself.

4.2.3.3 Church Road[65]

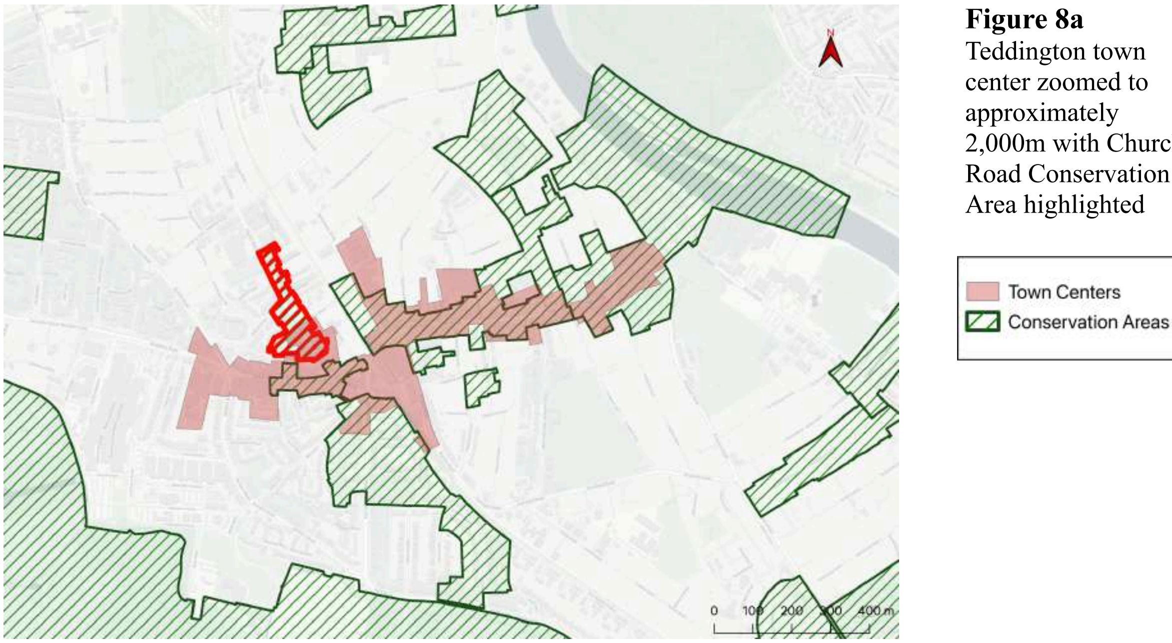


Figure 8a
Teddington town center zoomed to approximately 2,000m with Church Road Conservation Area highlighted

Source: Author's own diagram. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

[65]Urban Design and Conservation et al., *Church Road: Conservation Area No. 85*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2021).

b) Choice at 800m

c) Choice at 3,000m

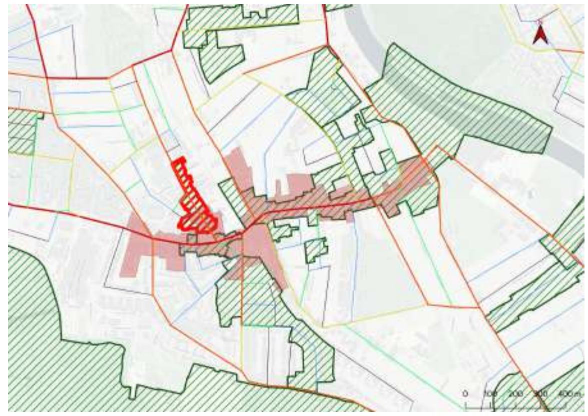
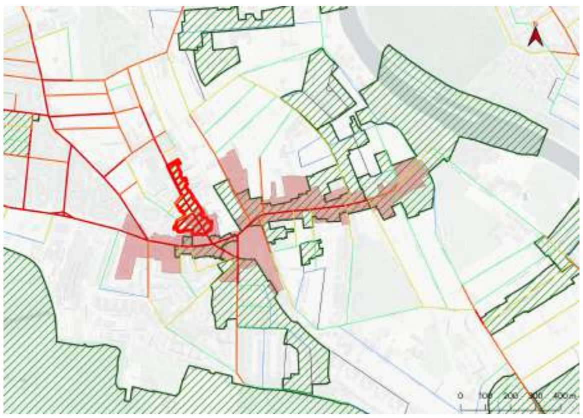
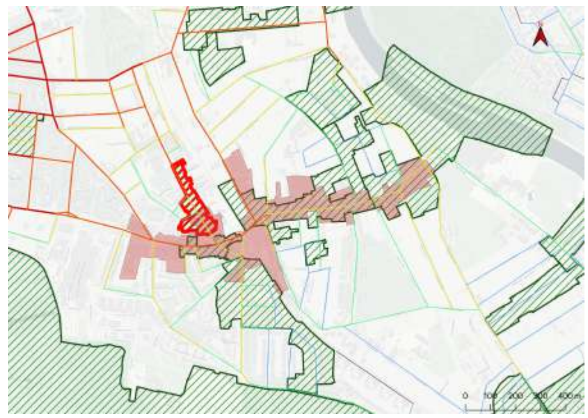


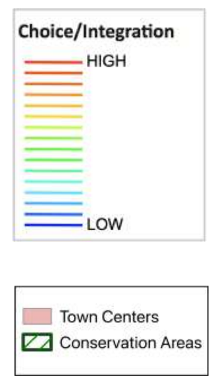
Figure 8b-g
Space syntax model of choice, integration, and combination, at radii 800m (left) and 3,000m (right) in the Church Road Conservation Area study area in sequential order (top to bottom).

d) Integration at 800m



e) Integration at 3,000m





f) Combination of choice and integration at 800m

g) Combination of choice and integration at 3,000m

Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

The Church Road Conservation Area lies just north of where it meets Broad Street. The conservation area only covers a fraction of Church Road despite the relative spatial importance visible along the entirety of its length to where it meets Shacklegate Road.

The language of the conservation area appraisal mirrors the spatial language used by Froy[66] when designating a hierarchy of street segments, stating that Church Road has a long history of commercial importance as a foreground segment as opposed to an auxiliary background segment. Its built structures are street-facing with the capacity for retail or other live activities. However, this use has not been sustained through to the present day and the appraisal fails to acknowledge the commercial importance of this road as a feeder to the high streets to the south and as a place of spatial capacity, as demonstrated by its strong co-presence values at both scales of movement. The small grid to the west of Church Road contains a number of non-residential uses lying outside of the bounds of the town center and any conservation area (figure 8h). This supports the notion posited by Vaughan et al. that high-activity roads such as Church Road, Broad Street, and High Street are supported by spatial morphologies that allow for a wider network of non-residential uses, increasing journeys of multiple scales and purposes.

The importance of Church Road has likely been amplified by the railway, which has historically prevented connectivity to the street segments lying to the west. This historical spatial relationship and its resulting effects on present-day movement dynamics are further unrecognized in the area appraisal.

[66] Froy, *Rebuilding Urban Complexity*.

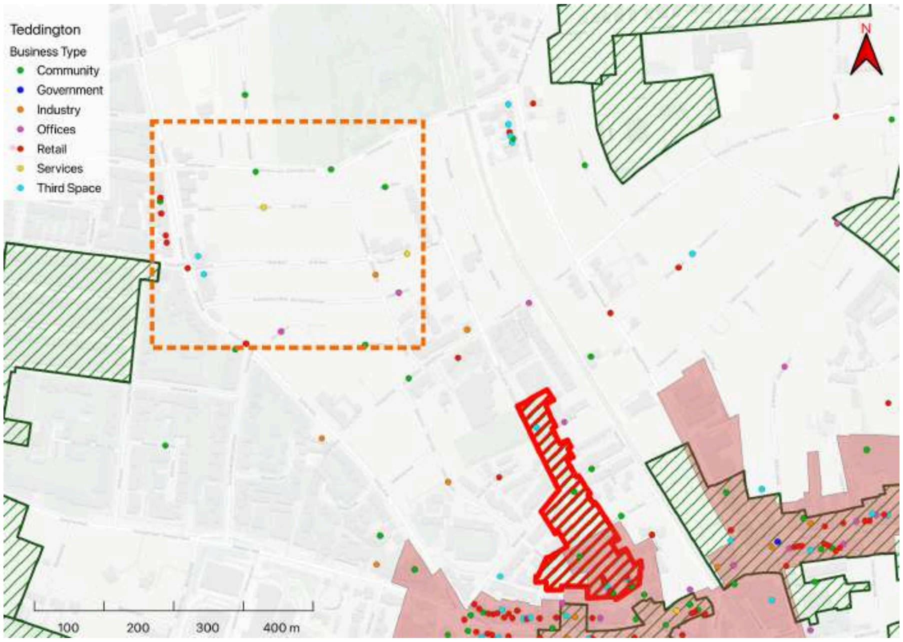


Figure 8h
Sample of non-residential land use distribution to the west of the northern segment of Church Road

Source: Author's own diagram. Land use data extracted via Google Maps Scraper (see https://console.apify.com/actors/nwua9Gu5YrADL7ZDj/information/latest/readme, by Compass, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

4.2.3.4 Teddington Lock[67]

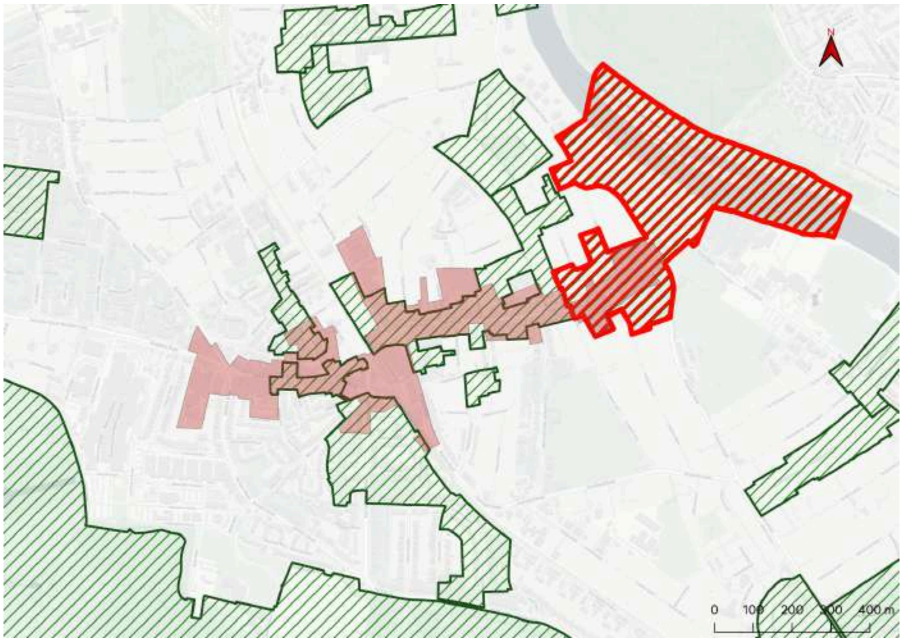


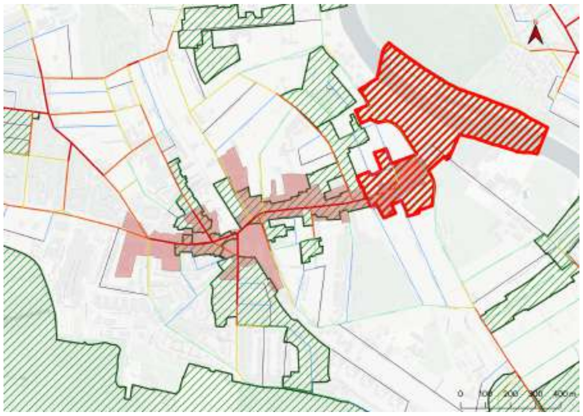
Figure 9a
Teddington town center zoomed to approximately 2,000m with Teddington Lock Conservation Area highlighted

Town Centers
Conservation Areas

Source: Author's own diagram. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

[67] Royal Borough of Richmond on Thames, *Teddington Lock: Conservation Area Appraisal Conservation Area No. 27* (Environment, Sustainability, Culture and Sports Committee, 2023).

b) Choice at 800m



c) Choice at 3,000m

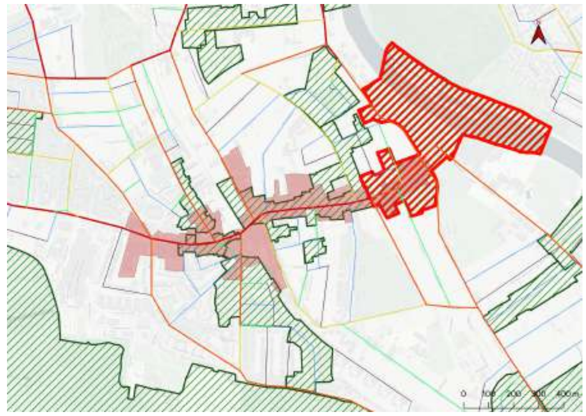
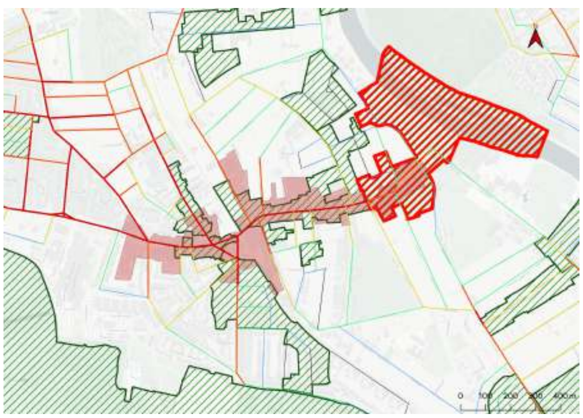
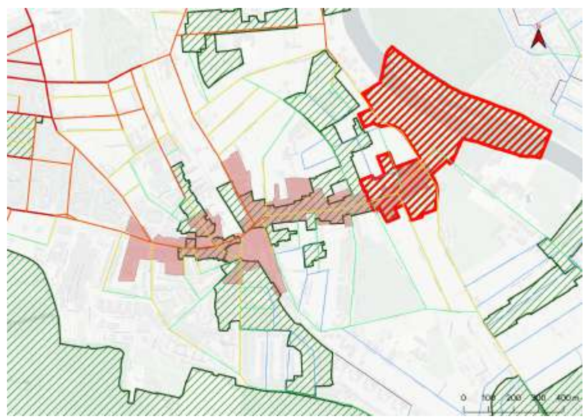


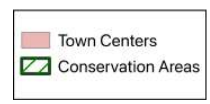
Figure 9b-g
Space syntax model of choice, integration, and combination, at radius 800m (left) and 3,000m (right) in the Teddington Lock Conservation Area study area in sequential order (top to bottom).

d) Integration at 800m

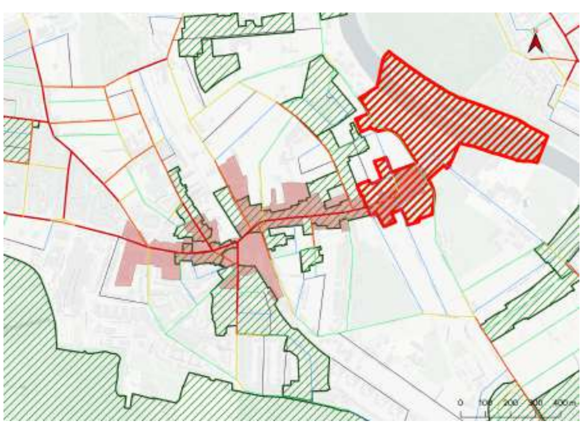


e) Integration at 3,000m

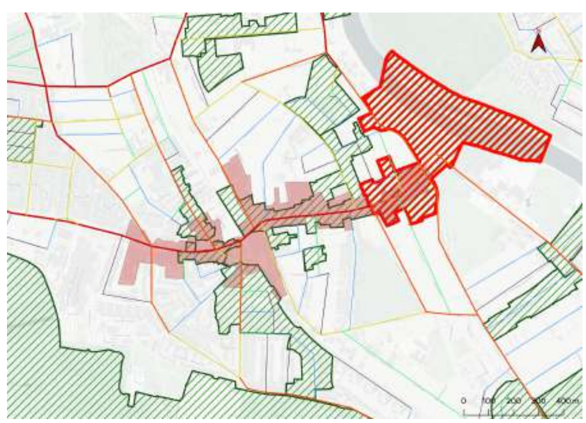




f) Combination of choice and integration at 800m



g) Combination of choice and integration at 3,000m



Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

The Teddington Lock Conservation Area primarily exists to designate the historic canal lock and weir that lie to the east of Teddington, but it also includes the easternmost part of High Street. This is another example of fragmentation that exists in what could be a unified conservation management plan for Teddington High Street.

It is worth noting that the spatial analysis presented here is limited by the fact that the public dataset of English roads on which it is based does not include the pedestrian bridge crossing the lock, which would likely change the results of the quantitative analysis and subsequent maps.

The conservation area sits on parts of the north-south routes of Twickenham Road and Broom Road that demonstrate high choice at a radius of 3,000m (figure 9c), again showing the importance placed on the long roads that connect Twickenham, Teddington, and Kingston.

The conservation area appraisal is heavily focused on the architectural and historic character of the lock and its visual setting, and makes no mention of the conservation of the section of the high street for which it is responsible.

Given that the dataset lacks the pedestrian route that links Teddington to the other side of the river north of Kingston, it functionally demonstrates the effect on connectivity in the areas that would otherwise be more integrated. The street segments of Saint Alban Gardens, Twickenham Road, and Manor Road lying just north of the high street show very low integration values suggesting their relative spatial isolation. It would be valuable to explore how co-presence and connectivity are affected on either side of the bridge should this segment be included in a future dataset, and to better quantify this area's importance for non-residential activity in Teddington.

4.2.3.5 Park Road[68]

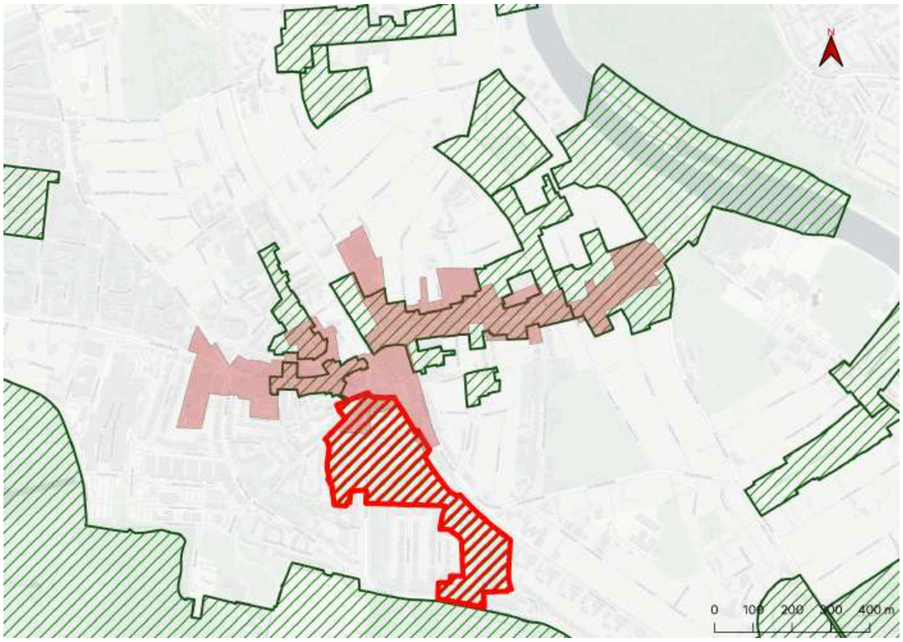
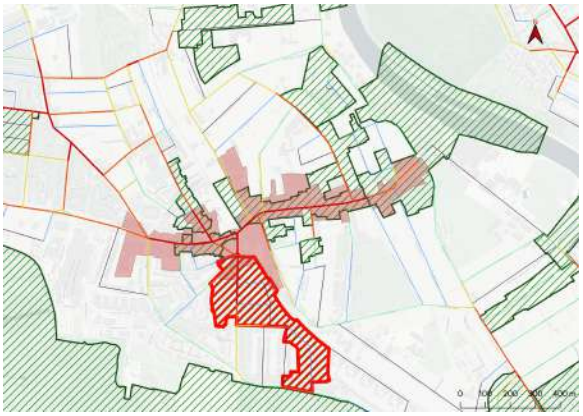


Figure 10a
Teddington town center zoomed to approximately 2,000m with Park Road Conservation Area highlighted

Source: Author's own diagram. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

[68] Urban Design and Conservation, *Character Appraisal and Management Plan: Conservation Area - Park Road No. 22*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2006).

b) Choice at 800m



c) Choice at 3,000m

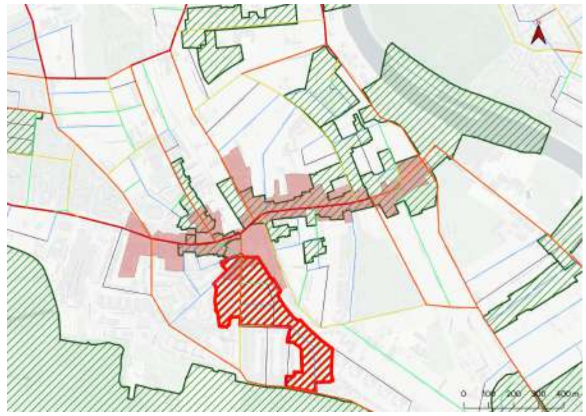
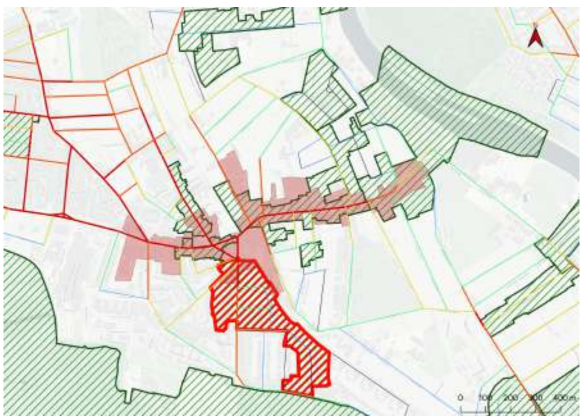
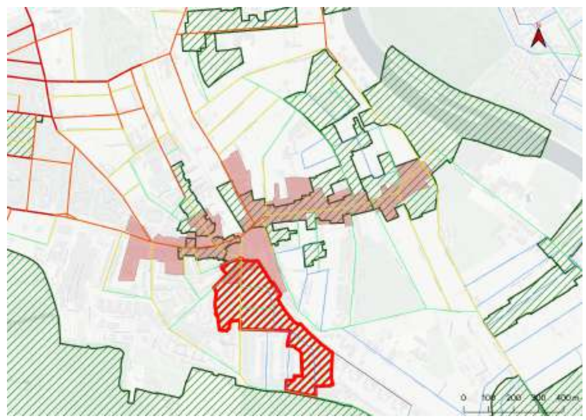


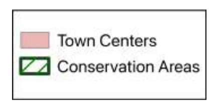
Figure 10b-g Space syntax model of choice, integration, and combination, at radius 800m (left) and 3000m (right) in the Park Road Conservation Area study area in sequential order (top to bottom).

d) Integration at 800m

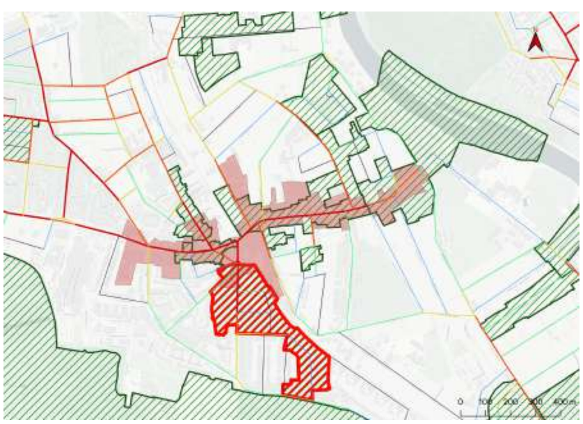


e) Integration at 3,000m

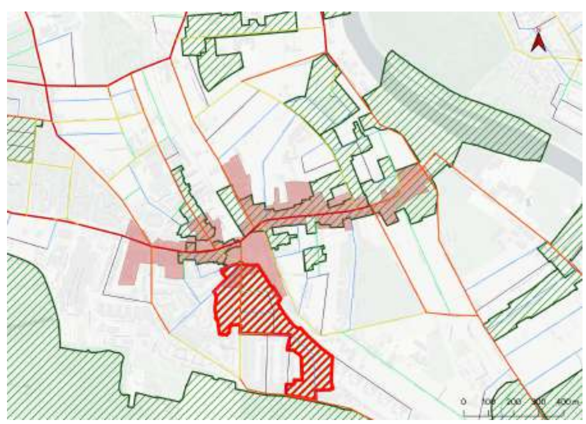




f) Combination of choice and integration at 800m



g) Combination of choice and integration at 3,000m



Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

The Park Road Conservation Area appraisal was one of the management plans with the most recognition of spatial relationships. The appraisal cites the importance of Park Road as an important connective route within the Teddington Town Center. This is supported by language choices such as "entrance," "lead to," and "tightly knit grid," descriptors often absent from conservation area appraisals in the same geographic region.

That said, the conservation area boundary only covers half of Park Road, which demonstrates relatively high and consistent integration along its entire length. However, the segment within the conservation area boundary does demonstrate higher levels of spatial potential, especially at the 800m radius (figure 10f). This suggests some alignment, by design or otherwise, between the boundary and the areas of greatest spatial importance.

The area shows much lower scores at a radius of 3,000m (figures 10c, 10e, 10g), suggesting that it is unlikely to see activity from journeys of a larger scale. This inability to foster urban activity at multiple scales presents a risk for the resiliency of this area and its ability to foster resiliency in the areas it connects. For example, at 3,000m the relative lack of spatial complexity is in contrast to the comparatively high importance of Queen's Road, whose route circumvents the high street altogether.

4.2.3.6 Other Area Notes

The conservation areas described above are those with the most direct relationship to the town center and serve as a valuable academic foundation for understanding how the entire sample of conservation area appraisals was scored. This section marks a handful of smaller examples of spatial alignment or misalignment within additional conservation areas.

The Fieldend Conservation Area,[69] despite its small size and limitation to a single estate, contains more descriptive and qualitative language than most of the other conservation areas regarding spatial relationships.

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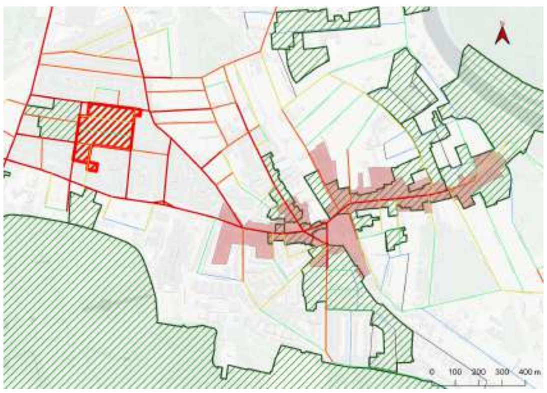
[69] Urban Design and Conservation, *Character Appraisal and Management Plan: Conservation Area - Fieldend No. 71*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2009).

The area around Royal Road[70] is better integrated at both movement scales than the town center proper and demonstrates a high level of spatial co-presence at multiple scales on street segments lying along the conservation area boundaries (figures 11a-d). This may be attributed to the fact that this area is less strictly bound by Bushy Park Gardens. Despite this, the conservation area appraisal for Royal Road has few references to these morphological realities and how they could affect the intended use of the area. Additionally, its relative connectivity is almost certainly related to the vitality of the town center, as it functions as a statistically likely journey "end point," many of which track through the high street.

a) Integration at 800m



b) Integration at 3,000m

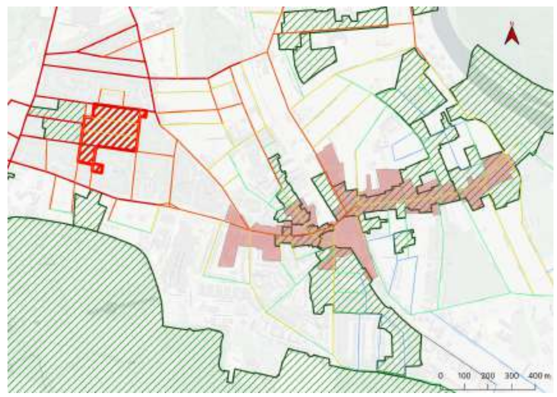
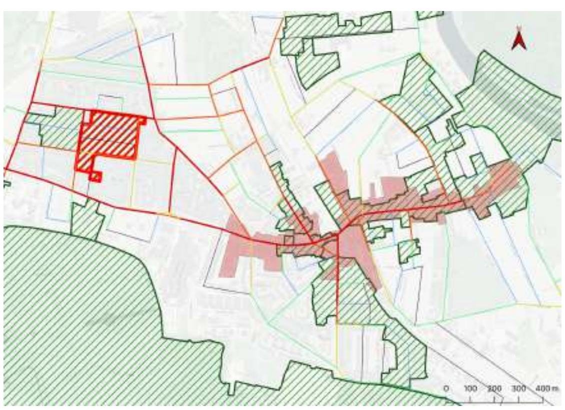
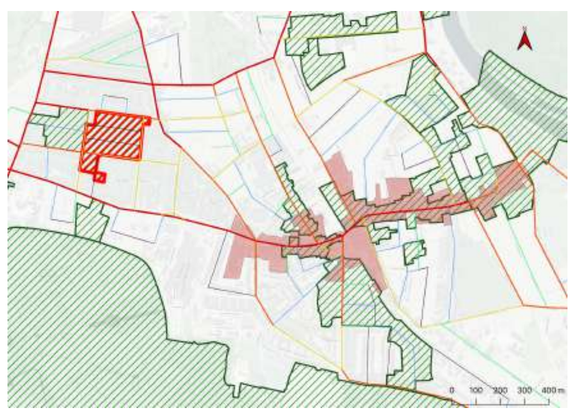


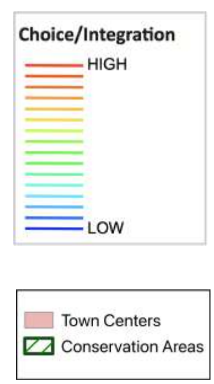
Figure 11a-d
Space syntax model, integration, and combination, at radii 800m (left) and 3,000m (right) in the Royal Road Conservation Area study area in sequential order (top to bottom).

c) Combination of choice and integration at 800m



d) Combination of choice and integration at 3,000m



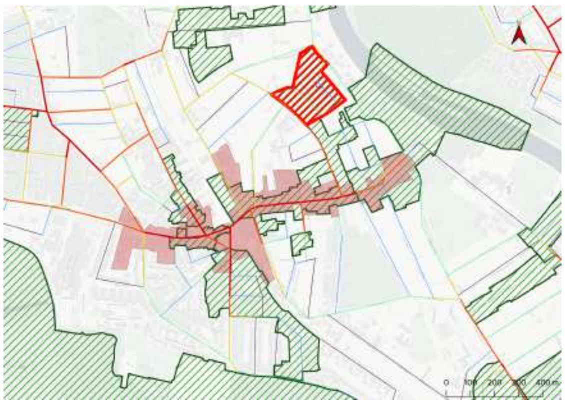


Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

[70] Urban Design and Conservation et al., *Royal Road: Conservation Area No. 81*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2021).

The Grove[71] Conservation Area appraisal describes its relative spatial isolation and relates this back to the historical ideologies upon which it was built. The lack of integration here has an interesting effect on through-movement, where north-south movement favors the route to the west or east of The Grove depending on journey length (figures 12a-d).[72]

a) Choice at 800m



b) Choice at 3,000m

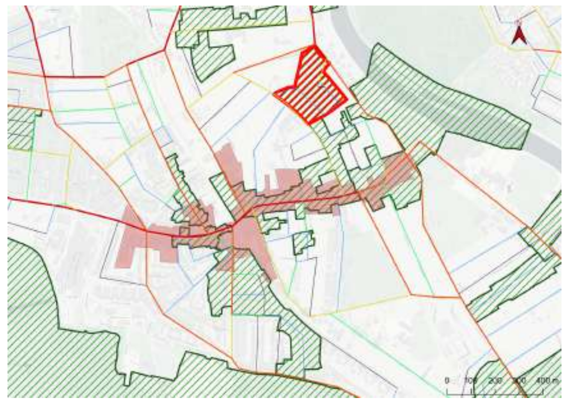
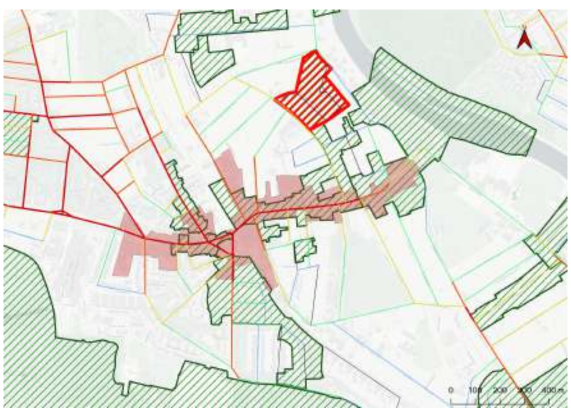
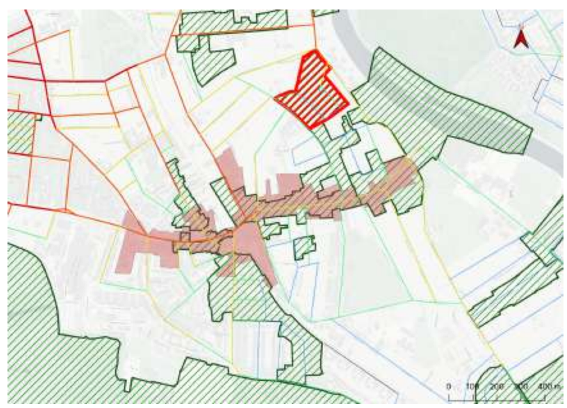


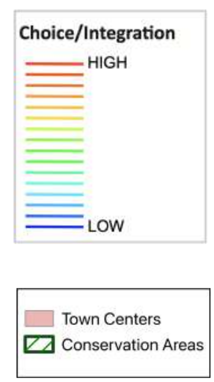
Figure 12a-d
Space syntax model of choice and integration at radii 800m (left) and 3,000m (right) in the The Grove Conservation Area study area in sequential order (top to bottom).

c) Integration at 800m



d) Integration at 3,000m





Source: Author's own diagram. Street network extracted from Space Syntax OpenMapping (see https://spacesyntax-openmapping.netlify.app/#13/51.4338/-0.3365, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

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[71] Urban Design and Conservation, *Character Appraisal and Management Plan: Conservation Area - The Grove No. 26*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2006).

[72] In the visualizations for both scales of integration one can see very few yellow, orange, or red lines indicating the area does not have a high degree of relative connection or "closeness." Perhaps related to this, choice is subsequently affected. At 800m (figure 12a) the road to the west of The Grove is orange, suggesting that for shorter journeys, one is more likely to use this as a through-route. At 3,000m, (12b) the road to east of The Grove is orange-red, suggesting this is the more "popular" route for longer journeys. This road to the east is also most likely the fastest route from Twickenham to the north and Kingston to the south, allowing one to avoid Teddington altogether on such trips.

Lastly, Field Lane intersects with Blackmore's Grove,[73] a small conservation area in Teddington that demonstrates the potential for co-presence at the smaller scale (figure 13), yet currently has no commercial activity. While this thesis does not suggest rezoning, this is an example of a spatial data point that could assist in these decisions going forward.

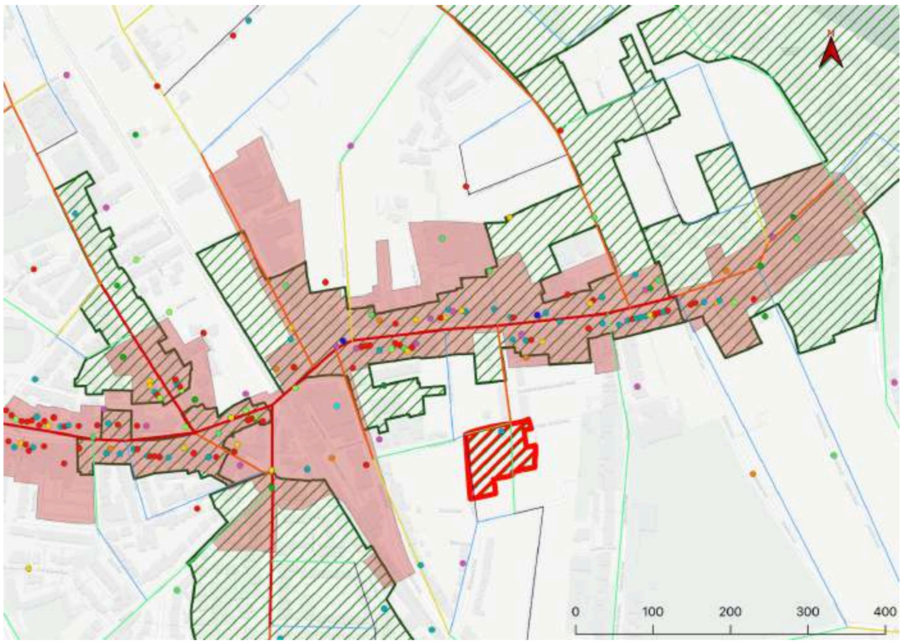
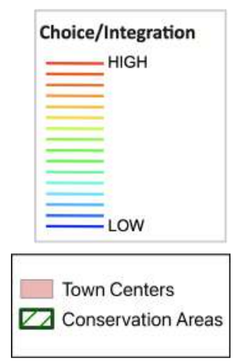


Figure 13
Space syntax model of the combination of choice and integration (co-presence) at 800m and sample non-residential land use distribution with a focus on Blackmore's Grove Conservation Area



Source: Author's own diagram. Land use data extracted via Google Maps Scraper (see https://console.apify.com/actors/nwua9Gu5YrADL7ZDj/information/latest/readme, by Compass, accessed [02/2026]), and amended by author. Local boundaries are derived from the December 2019 clipped boundaries dataset downloaded from www.data.london.gov.uk. Contains public sector information licensed under the Open Government Licence v3.0

5. Discussion & Conclusion

These results, although limited to two case studies, clearly represent a gap between the logic defining conservation area assessment and spatial network analysis. While a correlation exists between conservation areas and the existence of co-presence, this more likely reflects a consequence of overlapping priorities (i.e. a High Street can both be historical and an important place of urban activity), rather than a purposeful alignment. Generally, conservation area boundaries do not account for the area's

[73] Urban Design and Conservation, *Character Appraisal and Management Plan: Conservation Area - Blackmore's Grove No. 39*, Conservation Area Appraisal (London Borough of Richmond on Thames, 2006).

position within the larger network, nor do they account for how a conservation area—especially one designated to preserve a degree of urban activity—depends on the effects of the surrounding network of its setting. Appraisals rarely mentioned neighboring conservation areas, streets outside designated boundaries, or the patterns and capacity of movement in and through these areas.

Qualitative observations of the local plans of Croydon and Richmond on Thames suggest that conservation areas are mostly intended to restrict development. The results here demonstrate an opportunity for a shift in mindset from "restrictive" to "protective," where the latter is concerned with a continuation and advancement of what is working within areas of historic interest. Conservation policy is right to focus on the aesthetic appearance and intangible characters of neighborhoods, but the spatial-functional aspects of these neighborhoods are also worthy of consideration when defining heritage. This is best outlined by Dr. Kayvan Karimi's analysis of the transformation of the historic city over time:

> *"…a foundation for urban conservation is not a rigid and museum-like preservation. A successful conservation has to give way to adaptation and new elements as well. But the important point is how the city can be conserved without being frozen. A very important step in this direction is protecting the unique spatial system of the old cities, or the 'spatial spirit', which creates the relationships among the components and functions of the urban system."*[74]

This thesis also demonstrates a way in which space syntax can be used to identify localized spatial relationships between areas. It was used here to convey zones of high and low spatial potential as derived from spatial complexity, each of which may require different actions from policy makers and planners in the pursuit of different goals.

Additionally, the maps emphasize the fragmentation of conserved space, showing how difficult it is, for example, to truly attribute a single road to a single conservation area. In reality, street segments

[74] Kayvan Karimi, "Urban Conservation and Spatial Transformation: Preserving the Fragments or Maintaining the \`spatial Spirit'," *URBAN DESIGN International* (UK) 5 (2000): 231, https://doi.org/10.1057/palgrave.udi.9000012.

often overflow the boundaries of the conservation areas they are assigned to, and their spatial network effects can extend to neighboring areas and beyond.

South Norwood and Teddington are both defined as district centers by the London Plan which, although servicing a smaller area than metropolitan and major centers, still have significant commercial roles, nighttime economy goals, and resident, visitor, and business functions.[75] To accomplish this, these towns must be well integrated, have high spatial potential and complexity, and build upon their historic landscape and functions. The language of conservation area appraisals should be responsive to these goals; the study largely demonstrates that, currently, they do not.

What space syntax also reveals in this research are markedly lower co-presence scores at the larger movement scale. Given the intended area of influence district centers are meant to serve, this suggests that the spatial potential of some town centers is too weak to achieve planning goals that include these areas in medium to large scale movement networks.

The results suggest that urban vitality in conservation areas may be at risk, and that conservation area policy may not be equipped to handle these threats. For example, many of the visualizations do show some alignment between conservation area boundaries and areas of probabilistically high urban activity, which to some might seem sufficient. But this is a false tautology: high streets do not feature high levels of both choice and integration simply because they are high streets. It is rather a result of the morphological properties that produce high choice and integration that provide the necessary substrate for the high street's existence. As such, it is relevant to contextualize this spatial complexity in descriptions of these conservation areas. Because the high street is one street segment within a larger network, it is important to document that network co-presence is influenced by street segments that exist *outside* of the conservation area boundary. So while conservation areas may restrict change and conserve within their borders, change outside of those boundaries can still drastically affect the vitality of these areas. This

[75] Greater London Authority, "The London Plan: The Spatial Development Strategy for Greater London," 82–83, 92–94; Special Planning Service London Borough of Croydon, "Croydon Local Plan 2018 (Revised 2021)", Policy SP1; Richmond Upon Thames Local Plan: "The Best for Our Borough" (2024 to 2039), 48.

thesis argues that the definition of setting as provided by the NPPF is flexible enough to accommodate these descriptions.

Although limited in its ability to prove this, the research could be furthered by experiments in which streets segments within a chosen network radius are added or removed to observe the effect on choice and integration on street segments of interest.

Ultimately, none of the conservation area appraisals in this study identified all of the aspects of spatial complexity that support urban vitality. These aspects include the density of the urban grid, the relationship of the built environment to the street network, physical connectivity and movement capacity, and a rich mix of non-residential uses. Although some appraisals describe some of these aspects, space syntax is a useful tool in visualizing and mapping where these aspects exist and how they relate to the functioning of the area being appraised.

This thesis argues that urban vitality in historic areas of suburban London depends not only on the conservation of the aesthetic and cultural character of these areas, but also on their historic spatial networks and the movement of people through them both in the past and present.

Previous research has strongly suggested that the vitality of neighborhood centers and high streets is deeply intertwined to the physical characteristics of the built environment that allow for a rich and physically expansive mix of uses to sustain movement and activity over time. Numerous suburban London high streets lie within dedicated conservation areas, and their socio-economic role is as much a factor in their designation as their physical character. If this supporting research is accepted, it would be prudent to include descriptions of the network properties of space within conservation area appraisals.

Two representative case studies in South Norwood and Teddington show that conservation area appraisals do not—nor do their boundaries, stated boundary logic, or descriptions of setting—sufficiently account for or conserve swaths of space that are comprehensive enough to include the requisite network effects supporting urban vitality. This is demonstrated through thorough qualitative analysis of national policy language, local plan language, and individual conservation area appraisals in both Croydon and Richmond on Thames. This thesis also provides numerous maps overlaying conservation area and town center boundaries with novel segment angular analysis via space syntax methodology. The resulting visualizations further emphasize the gaps in alignment between boundary-based designations and areas with the spatial potential for vital urban activity.

However, this thesis cannot conclude on the effectiveness of current conservation policy in suburban London as it originally set out to do—it is a matter of fact that the town centers included in this study still exist and remain active—but it has illuminated discrepancies within the current expression of conservation policy and the potential benefit of spatial analysis. To steward these historic areas into an ever-growing and increasingly intertwined network of semi-urban and urban centers, spatial network analysis becomes an essential tool of heritage-based policy. Provided here is the beginning of a toolkit for incorporating spatial analysis into conservation area assessments. Nevertheless, it remains unclear whether spatial strengths and weaknesses should be addressed by conservation area appraisals themselves or by new supplementary guidance. This research could be carried further by longitudinal studies of physical change in the chosen case studies and beyond, more thorough surveys of land use patterns over space and time, experiments in physical intervention and network effects, and expansions to additional case studies both within and outside Greater London.

Urban space is a network of buildings, a network of streets, and networks of people. To not engage with these frameworks when describing the qualities and value of space is at odds with its actual functioning. Delineations of space that ignore these qualities risk perpetuating the fragmentation of space that unified frameworks such as the Historic Urban Landscape seek to overcome.

This has implications not just at the local London level, but globally, wherever historic areas at the heart of communities face threats of gentrification, isolation, dissolution, and destruction. It is paramount that the historical networks underpinning these urban centers be understood now, so that the relevant morphological properties and critical urban complexity can be properly conserved or rebuilt if necessary.

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